

FRM 二级 学霸笔记

Liquidity and Treasury Risk Measurement and Management

《流动性和司库风险测量与管理》

Chapter 1 Liquidity Risk(☆☆☆)	4
1. 概念介绍	4
2. Transaction liquidity risk 交易流动性风险	4
3. Funding liquidity risk 融资流动性风险	8
4. Liquidity black holes 流动性黑洞	11
Chapter 2 Liquidity and Leverage(☆☆☆)	13
1. Bank 的 Funding liquidity risk 来源	13
2. 不同金融机构面临的挑战	14
3. 抵押品市场	15
4. 杠杆 Leverage	16
Chapter 3 Early Warning Indicators 预警指标(☆)	20
1. 简介	20
2. Framework Components: M.E.R.I.T	20
3. Key Supervisory Guidelines	20
Chapter 4 The Investment Function in Financial-Services Management(☆☆☆)	23
1. Invest Instruments	23
2. Factors affecting choice of investment securities	25
3. Investment Maturity Strategies 投资期限策略	27
4. Maturity Management Tools 期限管理工具	29
Chapter 5 Liquidity and Reserves Management: Strategies and Policies(☆☆☆)	32
1. Net Liquidity and Strategies	32
2. Estimating Liquidity Needs 估计流动性需求	34
3. Legal Reserve 法定准备金	39
Chapter 6 Intraday Liquidity Risk Management(☆☆)	41
1. Uses and sources of intraday liquidity	41
2. Governance	45
3. Measurement of Intraday Liquidity	45
Chapter 7 Monitoring Liquidity(☆☆☆)	47
1. Taxonomy of the cash flows 现金流的分类	47
2. Liquidity Option 流动性期权	47
3. The new definition of Liquidity risk	48
4. Quantitative Liquidity risk Measures	49
5. The term structure of expected liquidity	58
6. Cash flows at risk and the term structure of liquidity at risk	59
Chapter 8 The Failure Mechanics of Dealer Banks(☆)	60
1. Large Dealer Banks 业务	60

2. Failure Mechanics of Dealer Bank	60
3. Policy Responses 政策回应	60
Chapter 9 Liquidity Stress Testing 流动性压力测试(☆)	62
1. Contingent Liquidity	62
2. Estimating Contingent Liquidity	62
3. Design of the model	63
Chapter 10 Liquidity Risk Reporting and Stress Testing(☆)	65
Chapter 11 Contingency Funding Planning 应急融资计划(☆☆)	67
1. 基本概念	67
2. Key Design Considerations	67
3. Framework and Building Blocks	67
Chapter 12 Managing and Pricing Deposit Services(☆☆☆)	69
1. Types of Deposits	69
2. Pricing Different Types of Deposits	70
3. Challenges of Deposit Offering	71
Chapter 13 Managing Non deposit Liabilities(☆☆)	73
1. Sources of Nondeposits	73
2. Chooses of Nondeposits	75
3. Costs of Nondeposits	76
Chapter 14 Repurchase Agreements and Financing(☆☆☆)	78
1. 回购基本概念	78
2. 回购的原因	79
3. Repos During the Credit Crisis	80
4. Collateral in Repo Transactions	80
5. Financing Value of A Bond Trade Special	81
Chapter 15 Liquidity Transfer Pricing 流动性转移定价(☆☆☆)	82
1. LTP Introduction	82
2. Governing LTP	82
3. 实施 LTP 面临的挑战	83
4. 风险定价	84
5. Contingent Liquidity Risk Pricing 或有流动性风险定价	87
Chapter 16 The US Dollar Shortage in Global Banking and the International Policy Response(☆)	90
1. 全球金融危机期间美元短缺的原因	90
2. Maturity Mismatch&Currency Mismatch	90
3. The international policy response	91
Chapter 17 Covered Interest Parity Lost 利率平价理论的失衡(☆)	93
1. 基本概念: Covered Interest Parity(CIP)抛补利率平价	93
2. FX Swaps and Cross-currency Swaps	93
3. Causes of CIP Violations	94
Chapter 18 Risk Management for Changing Interest Rates(☆☆☆)	96
1. 基本概念	96
2. Interest-sensitive Gap Management	96
3. Duration Gap Management	99

Chapter 19 Illiquid Assets 非流动性资产(☆☆)	101
1. Illiquid Asset Markets	101
2. The Source of Illiquidity Market Imperfections	101
3. Biases on Returns for Illiquid Assets	101
4. 获得流动性风险溢价的方法	103
5. Portfolio Choice Decisions	104

Chapter 1 Liquidity Risk(☆☆☆)

【Chapter1 和 2 都是关于流动性风险基本概念，但是选自两个人的书籍，所以会出现相同的概念有不同的解释，我这边尽量就相关内容进行了合并梳理】

1. 概念介绍

- Solvency 偿付能力 vs. liquidity 流动性
 - ✓ **Solvency: assets more than liabilities (equity is positive).** 【侧重长期】
 - ✓ **Liquidity: ability of a company to make cash payments as they become due.** 【侧重短期】
 - ✓ Solvency 和 liquidity 并不完全相同：
 - 偿付能力良好的金融机构（即有正资产净值）也可能会因为流动性问题而破产。例如，后面介绍的北岩银行。
- 流动性分为：
 - ✓ **Transactions liquidity 交易流动性**, a property of assets or markets
 - Transaction liquidity is the property of an asset being easy to exchange for other assets. 【侧重资产负债表的资产端，关注的是资产能不能及时足额的变现】
 - ✓ **Funding liquidity 融资流动性**, which is more closely related to credit-worthiness.
 - Funding liquidity is the ability to finance assets continuously at an acceptable borrowing rate. 【侧重资产负债表的负债端，关注的是能不能及时足额的“借来钱”】
- 流动性风险分类：
 - ✓ Transaction liquidity risk 交易流动性风险
 - ✓ Funding liquidity risk 融资流动性风险
 - ✓ Systemic risk 系统性风险
 - 系统性风险 refers to the risk of a general impairment of the financial system. 【这种风险比较严重，整个金融体系出现了问题】
 - ✓ 这三种流动性风险在一定情况下，可以相互转化。

2. Transaction liquidity risk 交易流动性风险

1) 基本概念

- Transaction liquidity risk, 在原版书有些章节也写成 liquidity trading risk
- 也叫 market liquidity risk 市场流动性风险
- **Transaction liquidity risk** is the risk of moving the price of an asset **adversely**

in the act of buying or selling it.

- ✓ 交易流动性风险指在买卖资产时对其价格产生不利影响的风险。【想买买不到，想卖卖不掉】
- ✓ 在不大幅打折或者大幅提升价格时，就能够快速卖掉或者买到，这说明交易流动性风险小。
- ✓ An asset is said to be liquid if it is "near" or a good substitute for cash. 【cash 是流动性非常好的资产】
- ✓ An asset is said to have a liquidity premium(流动性溢价) if its price is lower and expected return higher because it isn't perfectly liquid. 【投资者投资一个流动性不好的资产，会要求一个补偿，所以所要求的 return 会高，那么对应的价格会低（风险大的东西，“不好”的东西，价格才会便宜）。这个补偿就叫作流动性溢价 liquidity premium】

● Causes of Transactions Liquidity Risk

- ✓ Transaction liquidity risk is ultimately due to the cost of searching for a counterparty, to the market institutions that assist in search, and to the cost of inducing someone else to hold a position.【交易流动性风险主要来自这些方面的成本：找到交易对手、使用中介服务以及提供交易激励的费用。这些成本会影响你能多快和多容易地完成交易，如果成本高或者交易困难，风险就会增加。】
 - ✧ Cost of trade processing.佣金
 - ✧ Inventory 库存 management by dealers.做市商的库存
 - The role of dealers is to provide trade immediacy to other market participants, including other dealers, by holding long or short inventories of the asset. 【做市商提供交易的即时性，但需持有库存并承担价格风险，赚 bid ask spread】
 - ✧ Adverse selection 逆向选择
 - Some traders may be better informed than others, that is, better situated to forecast the equilibrium price. 【交易对手信息不对称导致的风险】
 - ✧ Differences of opinion 意见分歧
 - Investors generally disagree about the "correct" price of an asset 【投资者对资产价格和信息解释的分歧，增加了寻找交易对手的难度。】

● Characteristics of Market Liquidity

- ✓ A standard set of characteristics of market liquidity, focusing primarily on asset liquidity, helps to understand the causes of illiquidity:
 - ✧ Tightness (也叫 width) 紧度 refers to the **cost** of a round-trip transaction, and is typically measured by the bid-ask spread and brokers' commissions. 【Tightness: 一笔往返交易的成本, 通常用 bid ask spread 和经纪人的佣金来衡量】
 - ✧ Depth 深度 describes how **large an order** it takes to move the market adversely. 【需要多大的订单才能对市场产生不利影响。Depth↑, 流动性↑】
 - ✧ Resiliency 弹性 is the **length of time** for which a lumpy order moves the market away from the equilibrium price. 【一个大订单将市场价格偏离均衡价格的时间长度。lumpy order 大订单】
- ✓ **Lack of liquidity** manifests itself in these observable:
 - ✧ Bid-ask spread 波动大.
 - ✧ Adverse price impact arises. 交易者自身活动对市场价格的影响。
 - ✧ Slippage 滑点 arises
 - the deterioration in the market price induced by the amount of time it takes to get a trade done. 【缩量下跌】

【市场流动性的这些特征(尤其是后两者)难以衡量, 使得实证研究复杂化。】

2) 交易流动性风险的计量

- 用 Bid-Offer Spread 衡量流动性风险
 - ✓ BID Price: dealer 的买价, 更低。“你”的卖价
 - ✓ OFFER Price: dealer 的卖价, 更高。“你”的买价
 - OFFER Price 也叫 ask price
 - Bid-Offer Spread, 有时也叫 Bid-Ask Spread
 - ✓ 两种形式:

Bid-ask spread (dollar 形式): $p = \text{offer price} - \text{bid price}$ 【高减低】

$$\text{Bid-ask spread}(\% \text{形式}): s = \frac{\text{offer price} - \text{bid price}}{\text{mid market price}}$$

$$\alpha = \text{mid market price} = \frac{\text{offer price} + \text{bid price}}{2}$$

- Cost of liquidation 清算成本: 清算某个资产需要花费的成本
 - ✓ In normal market:

$$\text{Cost of liquidation} = \sum_{i=1}^n \frac{1}{2} \times (s_i \alpha_i)$$

✓ In stressed market:

$$\text{Cost of liquidation} = \sum_{i=1}^n \frac{1}{2} \times (\mu_i + \lambda \sigma_i) \alpha_i$$

- ✧ I: 某个资产
- ✧ N: the number of positions
- ✧ α_i : the dollar value of the position in the instrument
- ✧ λ : 标准正态分布对应的关键值（单尾）；99%置信水平， $\lambda=2.326$ ；95%置信水平， $\lambda=1.645$
- ✧ μ_i : Bid-Offer Spread 的均值
- ✧ σ_i : Bid-Offer Spread 的标准差
- ✓ Cost of liquidation 越大→清算起来的成本越大→流动性风险越大

● Liquidity-adjusted VaR

【把市场风险和流动性风险结合在一起→经流动性调整的 VaR。这个知识点在《市场风险测量与管理》Chapter 6 中提了一下】

✓ In normal market:

$$\text{Liquidity - Adjusted VaR} = \text{VaR} + \sum_{i=1}^n \frac{1}{2} \times (s_i \alpha_i)$$

✓ In stressed market:

$$\text{Liquidity - Adjusted VaR} = \text{VaR} + \sum_{i=1}^n \frac{1}{2} \times (\mu_i + \lambda \sigma_i) \alpha_i$$

$$\text{✧ Spread risk factor: } \frac{1}{2} \times (\mu_i + \lambda \sigma_i)$$

✓ 关于 VaR→Adjusting VaR for liquidating

■ 潜在假设：σ 不变；VaR 独立；均匀清仓

$$\text{VaR} \times \sqrt{\frac{(1+T) \times (1+2T)}{6T}}$$

- T is the required for the orderly liquidation of a position. 【T 是清仓所用的时间】
- Square root rule is overstating VaR for different time horizons
- ✓ 例如：if T=4, the adjustment for overnight VaR is 1.3693, which is less than 2.

拓展：

✓ 在市场风险或者在一级学过 Square root rule 平方根法则【平方根

法则假设头寸在整个时间段内是持有的，但在实际操作中，头寸通常会在 T 天内逐步清算。】：

$$\text{VaR}_{T\text{-day}} = \sqrt{T} \times \text{VaR}_{1\text{-day}}$$

- ✓ 但是在平仓的时候，用 Square root rule 不太合适会高估，需要对流动性进行调整，即

$$\text{VaR} \times \sqrt{\frac{(1+T) \times (1+2T)}{6T}}$$

例如：

- ✓ 一个投资组合的 1 天 95% 置信水平的 VaR 为 100 万元。假设该投资组合可以在 4 个交易日内逐步清算。

- a. 如果用平方根法则来调整 VaR, $\text{VaR}_{4\text{-day}} = \sqrt{4} \times 100 = 200$ 。这意味着如果整个头寸在 4 天内持有，则 VaR 被估算为 200 万元。

★ 这种方法高估了风险，因为它假设整个投资组合在 4 天内的每一天都面临同样的风险。

- b. 实际上，头寸的清算是逐步进行的，每天面临的风险是逐步下降的：

★ 第一天，整个投资组合面临 100 万元的 VaR。

★ 第二天，假设在第一天清算了 1/4 的头寸。此时，剩下的头寸是原始头寸的 3/4。由于已经清算掉的 1/4 头寸不再面临风险，剩下的头寸的风险会相应减少。

★ 第三天，剩下的头寸进一步减少，风险也随之降低。

★ 直到第四天。

$$\begin{aligned} \text{VaR} \times \sqrt{\frac{(1+T) \times (1+2T)}{6T}} &= 100 \times \sqrt{\frac{(1+4) \times (1+2 \times 4)}{6 \times 4}} \\ &= 136.93 \end{aligned}$$

- ✓ 136.93 小于 200
✓ 平方根法则高估风险。

3. Funding liquidity risk 融资流动性风险

【Funding liquidity risk，在原版书有的章节也写成 liquidity funding risk】

1) 基本概念

- 也叫 Balance sheet risk 资产负债表风险
- Funding liquidity risk, This is the financial institution's **ability to meet its cash needs** as they arise. 【典型：挤兑。借不来钱】
- 资金流动性风险在借款人用短期负债资助长期资产时更高，这被称为成熟度不匹配。

2) 流动性的主要来源：【钱从哪里来】

- Holdings of cash and Treasury securities 持有现金和国债。流动性好，但是收益低
- The ability to liquidate trading book positions 交易账簿的清算。【交易账簿的资产，通常用于交易目的。可以用于筹集现金，但还取决于市场的流动性】
- The ability to borrow money at short notice 在短时间内借款的能力。在紧张市场条件下，银行可能面临更高的风险厌恶情绪，导致借款利率上升、贷款期限缩短或拒绝提供资金。
- The ability to offer favorable terms to attract retail and wholesale deposits at short notice 在短时间内，以优惠条件吸引零售和批发存款的能力。
- The ability to securitize assets (such as loans) at short notice 可以在短时间内将资产（如贷款）证券化
- Borrowings from the central bank 从中央银行借款。

3) 融资流动性风险的 3 个 case:

- Financial institutions that are solvent (i.e., have positive equity) and sometimes fail because of liquidity problems. 【solvent 良好，但因为流动性而出现了问题的金融机构】
- Northern Rock 北岩银行
 - ✓ 北岩银行是一家英国银行，主要从事抵押贷款业务。
 - ✓ 使用短期债务工具融资，但在 2007 年至 2009 年次贷危机后，投资者不愿向银行借贷，导致资金问题。银行向英格兰银行寻求资金援助，但随后爆发了大规模的银行挤兑，最终导致银行被国有化。
- Ashanti Goldfields 阿尚蒂金矿
 - ✓ Ashanti Goldfields 是西非的一家金矿公司，通过卖出黄金远期来对冲黄金价格下跌的风险。然而，15 家欧洲央行宣布限制黄金销售，导致黄金供应减少，需求增加，黄金价格上涨超过 25%，触发了远期合约的保证金，导致公司面临重大损失。
- Metallgesellschaft (MG)
 - ✓ 德国金属公司(MG)，是德国的一个工业集团。使用短期期货合约滚动对冲，然而随后油价下跌，导致期货头寸需要追加保证金，引发公司资金流出问题，最终导致超过 13 亿美元的损失。

4) Measuring Funding Liquidity Risk 衡量融资流动性风险

【关于融资流动性风险的计量，原版书这里给出的方法是 Asset-Liability Management(资产负债管理)，这里涉及的内容可能在其他的风险管理书籍里看得不太一样，和后面(chapter 5)介绍的资产负债管理关注点也不太一样，这里比

较简单，没关系，这里了解下就行】

- **Asset-Liability Management(ALM)【针对银行】**
 - ✓ 【个人更倾向于把 Asset-Liability Management 作为流动性管理的一种方法，而不是计量流动性的一种方法。】
 - ✓ Tracking and forecasting available cash and sources of funding on the one hand, and cash needs on the other.【跟踪和预测可用现金及资金来源，同时预测现金需求。】
 - ✓ Keeping certain ratios of ready cash and readily marketable securities to meet unusual demands by depositors and other short-term lenders.【维持一定比例的现金和易于变现的证券以应对非常规的资金需求】
 - ✓ 这种 ALM 的核心：预计好未来需要的流动性，留下一部分易于变现的资产，以应对未来的不时之需。
- **Funding Liquidity Management for Hedge Funds【针对对冲基金】**
 - 【对冲基金易受流动性撤回的影响，需监控各种流动性来源】
 - ✓ **Cash** provides unfettered liquidity
 - ✓ **Unpledged assets** 未抵押资产 (or assets "in the box") are unencumbered assets not currently used as collateral.
 - ✓ **Unused borrowing capacity** on pledged assets can be used to finance additional positions.【未使用的抵押资产借款能力】

5) Regulations

【巴塞尔协议 3 提出了两个流动性指标，和流动性管理的指引（17 principles）。】

- **Liquidity Coverage Ratio 流动性覆盖率（LCR）**
 - ✓ LCR: 优质流动性资产储备与未来 30 日的资金净流出量的比值，且该比值应大于等于 100%
$$LCR = \frac{\text{High-quality liquid assets}}{\text{Net cash outflows in a 30-day period}} \geq 100\%$$
 - ✓ High-quality liquid assets 优质流动性资产：无折扣的 HQLA 包括政府证券、存款；企业债务和权益资产则有 50% 的折扣。
 - ✓ Net cash outflows in a 30-day period 30 天净现金流出
 - ✓ 关注短期流动性问题
 - ✓ 无折扣的 HQLA 包括政府证券存款；企业债务和权益资产则有 50% 的折扣。
- **净稳定融资比率（net stable funding ratio, NSFR）**

- ✓ 关注长期流动性问题

$$NFSR = \frac{\text{可用的稳定资金 (Amount of stable funding, ASF)}}{\text{所需的稳定资金 (Required Amount of stable funding, RSF)}} \geq 100\%$$

- ✓ ASF: 通过将每类资金（资本、批发存款、零售存款等）乘以可用稳定资金（ASF）因子计算，反映其稳定性。
 - ❖ 例如: capital (ordinary shares), 100%; stable deposits, 90%; less stable deposits, 80%; ...
 - ✓ RSF: 根据需要融资的资产和表外项目计算。每类项目乘以所需稳定资金（RSF）因子，反映融资的持久性。
 - ❖ Examples: cash, 0%; mortgages, 65%; ...
- 关于流动性风险管理的原则，BIS 提供了一个框架（17 principles），总结如下：
- ✓ 全面性：涵盖了流动性风险管理的各个方面，包括建立风险容忍度、制定策略、监测风险暴露和定期进行压力测试。
 - ✓ 风险导向的方法：强调了将流动性风险管理 with 银行的业务战略和风险容忍度相一致的重要性。
 - ✓ 积极管理：银行要积极监测和控制流动性风险暴露，定期进行压力测试，并持有高质量的流动性资产储备。
 - ✓ 应急计划：制定正式的应急资金计划（CFP），以应对紧急情况下的流动性不足。包括策略、责任、启动程序，并定期测试以确保运营的稳健性。
 - ✓ 透明度：定期公开其流动性风险管理框架和状况的信息。
 - ✓ 监管监督：监管机构负责定期评估银行的流动性风险管理框架和状况，以确保对流动性压力具有抵御能力。在必要时，应该进行干预。

4. Liquidity black holes 流动性黑洞

- 有时也叫作 crowded exit(拥挤的退出)
- A **liquidity black hole 流动性黑洞** describes a situation where liquidity has dried up in a particular market because everyone wants to sell and no one wants to buy, or vice versa.
 - ✓ 流动性黑洞，指的是市场上只有卖家而没有买家，或者只有买家而没有卖家的流动性枯竭的现象。
- Negative feedback traders

- ✓ **Negative feedback traders** buy when prices fall and sell when prices rise.
【好现象，价格低→买；价格高→卖】
- ✓ In liquid markets, negative feedback traders dominate the trading .
 - 在负反馈交易者主导的市场中，资产价格过低→交易者买入→对资产需求增加→资产价格上升，恢复到合理水平。
 - 同样，当资产价格过高时→交易者会卖出→资产供应增加→价格下降，也使价格恢复到合理水平。
- Positive feedback traders
 - ✓ **Positive feedback traders** sell when prices fall and buy when prices rise.
【坏现象，价格低→卖；价格高→买】
 - ✓ When positive feedback traders dominate the trading, market prices are liable to be unstable and the market may become one-sided and illiquid.
 - 当正反馈交易者主导市场时，市场价格容易不稳定，市场变得单边、流动性不足。
 - 资产价格下跌→卖出→进一步推动了价格的下跌，更多的卖出。
 - 资产价格上涨→购买→进一步推动了价格的上涨，更多地购买。
 - ✓ 造成正反馈交易存在的原因：【这些因素导致了市场价格的不稳定和流动性问题，从而引发流动性黑洞的出现。】
 - 趋势交易 Trend trading: 在价格趋势上升时购买，在价格趋势下降时卖出。
 - 止损规则 Stop-loss rules: 当资产价格下跌到一定水平时，会自动卖出来限制损失
 - 动态对冲 Dynamic hedging:
 - Short option hedging→positive feedback
 - Long option hedging→negative feedback
 - 合成期权 Creating options synthetically
 - 保证金 Margins
 - 掠夺性交易 Predatory trading
 - 类似于长期资本管理公司(LTCM)的失败问题: short liquid bond and long illiquid bond

Chapter 2 Liquidity and Leverage(☆☆☆)

融资流动性风险，不仅商业银行会面临，其他金融机构也会面临。我们需重点关注商业银行的**融资流动性风险**。

1. Bank 的 Funding liquidity risk 来源

- The core function of a commercial bank is to take deposits and provide commercial and industrial loans to non-financial firms. 【商业银行的**核心功能**是吸收存款，发放贷款】
- In doing so, it carries out a **liquidity and maturity**, as well as a **credit transformation**.
- 【小背景：银行作为中介机构具备三大功能，包括信用转换（credit transformation）、期限转换（maturity transformation）和流动性转换（liquidity transformation）。这三大功能致使银行资产负债表非常脆弱，一旦出现信用危机，极容易面临被挤兑的风险。】
- 融资流动性风险来源：Maturity Transformation(期限转换)和 liquidity transformation 流动性转换带来的风险。【考纲要求重点掌握期限转换】

1) **Maturity Transformation(期限转换)**:

- ✓ **Maturity mismatch** 期限不匹配: borrow at short term to finance investments that require a longer time to become profitable. 【借短投长，储户的存款通常短期的多，而银行的贷款等通常长期的多】
 - Funding long-term assets with short-term debt exposes an intermediary to **rollover-risk** or even **cliff-risk**.用短期债务资金支持长期资产会使中介机构（这里通常指商业银行）面临滚动风险 rollover-risk，甚至 cliff-risk。
 - 如果银行无法及时或只能以不利条件重新融资，可能面临无法支付到期债务的风险，银行可能有流动性风险，甚至导致债务违约和财务危机。

2) **Liquidity transformation** 流动性转换:

- ✓ 流动性转换是期限转换的结果。利用流动性好的短期存款发放流动性差的长期贷款

3) **Credit transformation** 信用转换: 【原版书没有给出具体内容,这里属于补充知识】

✧ 银行以其信用替代了借款人的信用。

- 商业银行从储户吸收存款，再将其放贷给借款人。无论借款人是否按期偿还债务，储户都可以要求银行偿还本金和利息。

- To manage funding liquidity risk, bank using asset liability management (ALM) 资产负债管理.
 - ✓ ALM 资产负债管理的核心：确保银行在资产和负债之间保持平衡，最大程度地降低风险并确保满足资金需求。
 - ✓ 这包括对资产和负债的期限、流动性和利率风险进行管理。
- 2. 不同金融机构面临的挑战
 - 1) Commercial banks:
 - Bank run 银行挤兑：银行是部分准备金制度（fractional-reserve），如果存款人担心银行的流动性出现问题，抢先要求银行兑现。就会出现银行挤兑。
 - 部分准备金制度（fractional-reserve）：监管规定银行必须将一部分存款留下来，作为准备金，而剩下的部分资金才可以放贷或进行其他投资。
 - 2) Security firms:
 - Securities firms hold inventories of securities for sale, and finance them by borrowing at short term.
 - 2008 年 3 月，贝尔斯登公司的崩溃(The collapse of Bear Stearns)，贝尔斯登和其他大型 broker 一样，很大程度上依赖于短期借款。随着次贷危机的爆发，市场对贝尔斯登失去信心。投资者开始撤回对贝尔斯登的投资，同时，贝尔斯登持有的复杂金融工具的市场价值也大幅下跌，进一步削弱了其流动性。
 - 3) Money Market Mutual Funds (MMMFs)
 - MMMFs 通常由一家管理公司管理，并投资于短期、高信用质量的金融工具，如政府债券、短期企业债券等。
 - MMMFs 的资产价值会因市场变化而波动。如果资产价值下降，MMMFs 可能无法满足投资者的提取资金需求，进而面临流动性风险。
 - "breaking the buck" phenomenon: 是指 MMMFs 的净资产价值（NAV）小于每股名义价值 1 美元的现象。
 - ◆ 这种情况会损害投资者信心，导致更多的赎回请求，进一步加剧基金的流动性风险。
 - 4) Hedge funds:
 - 平衡投资者的提取需求、维护资本结构的稳定性，以及面对可能的短期资金压力。

3. 抵押品市场【这里可以结合《信用风险》科目中的“Chapter 13 CCR 管理-2. 抵押品”一起理解】

- **Markets for collateral** are formed when securities are used as collateral to obtain secured loans of cash or other securities.
 - ✓ 再质押 **rehypothecation** 或再抵押 **repledging** :Securities used as collateral "circulate 流通,"since the borrower of securities can typically lend them to another, a practice called **rehypothecation or repledging of collateral**.【再质押/再抵押指的是把抵押品再抵押给其他人】
- 抵押品市场让金融系统里的钱变得更容易借到、更便宜，显著提升了金融系统的流动性。
- 抵押品市场有两个主要目的：
 - ✓ 首先，they create the ability to **establish leveraged long and short positions** in securities.
 - ✓ 其次，抵押品市场**增强了公司的借款能力**。
- 抵押品市场的风险：
 - ✓ 抵押品市场的风险与其他杠杆头寸的风险类似，会有包括市场风险、信用风险和交易对手风险。
 - 借出和借入证券的双方都面临一些共同的风险，以及一些仅限于特定一方的风险。
 - ✓ The size and structure of collateral markets also contributes to **systemic risk**。抵押品市场规模庞大、交易复杂，一旦市场出现不利影响，可能会对整个金融体系产生连锁反应，从而增加系统性风险。
- Transactions in collateral market
 - A. Margin loans
 - B. Repurchase agreements
 - C. Securities lending
 - D. Total return swaps
- A. 保证金贷款（margin loans）
 - **保证金贷款**：可以理解为找 **broker** 借入资金，在市场上购买证券，而购买的这些证券作为抵押品，托管在 **broker**。
 - Margin lending is generally **short-term but rolled over automatically**.【保证金贷款一般为**短期贷款**，但会自动续期。】
 - 把购买的证券抵押给 **broker**，以“street name”方式保存

- ✓ **street name** 经纪人名义,指证券登记在 **broker** 名下, 而不是所有者的名下。
- ✓ 这简化了在贷款未及时偿还时扣押和出售证券的过程。
- ✓ 同时,也使得 **broker** 可以将这些证券用于其他用途,例如借给其他客户,在有担保的货币市场上借款等,此时这些证券叫作 **repledged securities**
- 保证金贷款,投资者需要递交初始保证金(**initial margin**)给 **broker**,美国监管机构通常设定初始保证金为 50%。
- 当抵押的证券价格大幅下跌时, **broker** 会要求投资者补充保证金。
- 杠杆【学完稍后的“杠杆 Leverage”,再看这个点】
 - ✓ 初始保证金(**initial margin**)就是杠杆中的 E。初始保证金为 50%, 即 $\text{Haircut}(h)=50\%$
 - 杠杆 $L=2(=1+\frac{D}{E}=1+50\%/50\%)$
 - 或者杠杆 $L=1/50\%$
- B. Repurchase agreement (Repo) 回购协议【详细见“Chapter 14 Repurchase Agreements and Financing”】
- C. 融券业务(**securities lending**) → 做空
 - 证券借贷是指一方(出借方)将其证券借给另一方(借入方),并收取一定的费用(通常称为回扣 **rebate**)。
 - ✓ 出借方:通常是大型机构投资者,如养老基金、保险公司等,他们将证券借给借入方。**继续享有证券的股息和利息。**
 - ✓ 借入方:通常是希望卖空这些证券的交易员或机构,他们借入证券后会将其卖出,希望在未来以更低的价格买回,从而获利。
 - 证券借贷的核心在于出借方通过借出证券赚取额外收入,而借入方通过借入证券实施卖空操作。从而增加了市场的有效性和流动性
- D. 总收益互换(**total return swap**)
 - 场外交易(OTC derivatives),用于进行股票的做空交易。
 - 一方支付固定费用,并获得指定股票头寸的总收益(这个收益是浮动的)

4. 杠杆 Leverage

- **Leverage ratio** = $\frac{A}{E} = \frac{D+E}{E} = 1 + \frac{D}{E}$
 - ✓ A: asset; D: debt 借来的钱; E: equity 自己的钱
 - ✓ $A=D+E$

- Leverage effect 杠杆效应:

- ✓ The leverage effect is the increase in equity returns that results from increasing leverage and is equal to the difference between the returns on the assets and cost of funding.

$$r_E = L * r_A - (L - 1)r_D = r_A + \frac{D}{E}(r_A - r_D) = r_D + L * (r_A - r_D)$$

- ✓ Effect of increasing leverage: $r_A - r_D$

r_A = return on assets

r_E : return on equity(ROE) 【本质上用自己的钱赚的收益率】

r_D : cost of debt

L = leverage ratio

【Tips:leverage effect 其实就是无税的 MM 理论 2。如果不太清楚 MM 理论的话,忽略这个 Tips。MM 理论在 FRM 里不是重点。】

- Haircut(h): 自己的钱(保证金), 通常带%

- ✓ 1-h: 借钱比例

$$\text{leverage} = \frac{1}{h}$$

- ✓ 随着资产价值的变化, 自己的钱不变(\$), 但是 Haircut 会变, leverage 也会变。
- ✓ margin loans, securities lending, 衍生品交易和 structured credit 中的 equity tranche 都自带杠杆

- Leveraged returns to housing(Assuming 10% price appreciation) 【房贷】

$$\text{Leverage} = 1 / \text{down payment}(\%)$$

- ✓ down payment 首付比例

- 原版书改编的例子: Suppose house values are expected to rise 10 percent a year,including, for simplicity, rental income and net of property maintenance costs. If the down payment of the house buyer is 20%,what is the annual rate of return to the homeowner by 6% loan rate?

$$\text{ROE} = \frac{\text{House price appreciation} - (1 - \text{Down payment})\text{Loan rate}}{\text{Down payment}}$$

- ✓ 【这个公式不用特意记忆, 主要知道这些专有名词含义和 ROE 的本质就行。这个公式, 在原版书中也是在例子中给出来的】
- ✓ House price appreciation 房价升值: 指的是房价上升, 这里的 10%指的是房价上升了 10%。比如如果原来房价是 100 元, 现在变成了 110, 赚了 10 元, 也就是 $10/100=10\%$

- ✓ Down payment 指的是首付比例。这里是首付 20%，剩下的 80% 是贷款
- ✓ Loan rate 贷款利率，借钱要给别人利息
- ✓ ROE 本质自己的本钱赚/亏了多少
 - 赚钱：赚的钱减去成本 = $10\% - 80\% \times 6\%$
 - 自己的钱：20%
 - ✓ $ROE = (10\% - 80\% \times 6\%) / 20\% = 26\%$
- Gross leverage 和 Net leverage
- ✓ Gross leverage 总杠杆 is defined as **the sum of all the asset values**, including cash generated by shorts or assets acquired with that cash, divided by **capital**.
【capital 资本，自己的钱】
 - 总杠杆反映了投资组合中所有资产相对于投资者自有资本的倍数

$$\text{Gross leverage} = (\text{多头头寸价值} + \text{空头头寸价值}) / \text{资本}$$
- ✓ Net leverage 净杠杆 is computed as the ratio of the **difference between the market values of the long and short positions** to the **capital**.
 - 净杠杆反映了投资者净持有资产的价值相对于其自有资本的倍数

$$\text{Net leverage} = (\text{多头头寸价值} - \text{空头头寸价值}) / \text{资本}$$
- ✓ 例子：
 - 假设投资者有 100 万的自有资本，并且进行了如下投资：
 - 直接购买了价值 150 万的资产。
 - 通过卖空股票获得 50 万的现金，并用这 50 万购买了其他资产。
 - 在这种情况下：
 - ✧ 多头头寸的市场价值：150 万。
 - ✧ 空头头寸的市场价值：50 万。
 - ✧ 资本：100 万。
 - ✧ 因此，总杠杆 = $(150 + 50) / 100 \text{ 万} = 2$ 。
 - ✧ 净杠杆 = $(150 \text{ 万} - 50 \text{ 万}) / 100 \text{ 万} = 1$ 。

6. Liquidity and System Risk

- Systemic risk, the risk of severe, widespread financial stress and intermediary failure, possibly including disruption of payment systems, is a function among other things of **economy-wide liquidity**.
- Funding Liquidity and Solvency
 - ✓ But a firm can be insolvent, yet able to continue for some time to roll over its debts, or may be funded largely by long-term debt, and thus not face

illiquidity.

- ✓ illiquidity can become insolvency if it is extreme enough
- Funding and Market Liquidity
 - ✓ A key mechanism linking funding and market liquidity is leverage.
 - 举例：Margin loan（杠杆），资产价格下降，可能触发 margin call，补充保证金，可能需要卖资产，导致资产价格进一步下跌，有可能导致进一步的 margin call。→ 恶化融资流动性风险和交易流动性
- Systemic Risk and the "Plumbing"
 - ✓ Disruptions in the payments, clearing, and settlements systems, often called the "plumbing" of the financial system, finally leads liquidity risk events become systemic risk events.

Chapter 3 Early Warning Indicators 预警指标(☆)

1. 简介

- Early Warning Indicators(EWIs 预警指标):类似于汽车仪表盘上的警示灯,用于提示银行流动性风险管理(liquidity risk management,LRM)的重要信息。
- EWI 作为流动性风险的早期检测系统,使银行能够及早发现潜在的流动性危机。

2. Framework Components: M.E.R.I.T

- EWI, 即早期预警指标, 在流动性风险管理过程中起着至关重要的作用。
- 1) Measures 度量指标: EWI 作为一种度量工具, 评估资产负债表结构、预测现金流量, 并评估未来的流动性状况。考虑了资产负债表内外、不同业务条件下的风险。
 - ✓ 包括内部和外部指标
 - ✓ 前瞻性: EWI 包括一系列领先指标、并具有足够的细粒度。
 - ✓ EWI 覆盖不同的时间范围, 包括日常、每周或每月的周期, 以适应银行的预测和业务运营。根据业务条件或监管要求, 某些指标甚至可能进行即时监测。
 - 2) Escalation 升级: 确保超出限制的情况被提升到适当的层级。设置不同级别的响应措施。
 - 3) Reporting 报告: 及时的 EWI 报告, 权衡覆盖范围和关键信息, 通常以仪表板形式呈现。
 - 4) Integrated systems 整合系统: 自动化和整合至关重要, 特别是随着 EWI 越来越多地包含一系列内部指标以及外部指标。
 - 5) Thresholds 阈值: 通常使用红黄绿灯系统表示指标相对于预定限制的表现。阈值应根据历史数据分析进行校准, 不至于太宽也不至于太窄。

3. Key Supervisory Guidelines

1) 美国国家银行监督管理局(OCC)-2012

- A bank should have EWIs that signal whether **embedded triggers in certain products** (i.e., call- able public debt, OTC derivatives transactions) are about to be breached, or whether contingent risks are likely to materialize.
- 银行应具备早期预警指标(EWIs), 提前识别潜在风险。早期预警指标可能包括:
 - ✓ A reluctance of traditional fund providers to continue funding at historic levels 传统资金提供者不愿意以历史水平继续提供资金

- ✓ Pending regulatory action (both formal and informal) or CAMELS component or composite rating downgrade(s) 有待采取的监管行动（正式和非正式），或者 CAMELS 或评级的下调
 - ✓ Widening of spreads on senior and subordinated debts, credit default swaps, and stock price declines 利差扩大
 - ✓ Difficulty in accessing long-term debt markets 难以进入长期债务市场
 - ✓ Reluctance of trust managers, money managers, public entities, and credit-sensitive funds providers to place funds 信托经理、货币经理、公共实体和信用敏感资金提供者不愿提供资金
 - ✓ Rising funding costs in an otherwise-stable market 资金成本上升
 - ✓ Counterparty resistance to off-balance-sheet products or increased margin requirements
 - ✓ The elimination of committed credit lines by counterparties 交易对手取消承诺的信贷额度
- 2) 巴塞尔银行监管委员会（BCBS）
- 有两个文件，分别是 2008 年的 Principles for Sound Liquidity Risk Management and Supervision 和 2012 年的 Monitoring Indicators for Intraday Liquidity
 - BCBS-2008
 - ✓ A bank should design a set of indicators to identify the emergence of increased risk or vulnerabilities in its liquidity risk position or potential funding needs.
 - ✓ 包括但不限于以下：
 - Early warning indicators can be qualitative or quantitative in nature and may include but are not limited to
 - Rapid asset growth, especially when funded with potentially volatile liabilities 快速资产增长，尤其是当其以潜在波动性负债融资时
 - Growing concentrations in assets or liabilities 资产或负债的集中程度增加
 - Increases in currency mismatches 货币错配的增加
 - Decrease of weighted average maturity of liabilities. 负债加权平均到期日的减少
 - Repeated incidents of positions approaching or breaching internal or

regulatory limits 反复出现或接近或突破内部或监管限制的持仓事件

- Negative trends or heightened risk associated with a particular product line 某种产品线出现不利趋势或风险加剧

- BCBS- 2012

- ✓ Intraday liquidity monitoring indicators include:

- Daily maximum liquidity requirement 日最大流动性需求
- Available intraday liquidity 可用的日内流动性
- Total payments 总支付额
- Time-specific and other critical obligations 特定时间和其他关键义务
- Value of customer payments made on behalf of financial institutions customers 代表金融机构客户进行的客户支付金额
- Intraday credit lines extended to financial institution customers 向金融机构客户提供的日内信贷额度
- Timing of intraday payments 日内支付的时机
- Intraday throughput 日内吞吐量

3) SR 10-6

- 2010 年美联储的 Federal Reserve (Supervision and Regulation [SR] 10-6) Guidelines
- Institution management should **monitor for potential liquidity stress events** by using early-warning indicators and event triggers. The institution should tailor these indicators to its specific liquidity risk profile.
- Early-warning signals may include, but are not limited to
 - ✓ Negative publicity concerning an asset class owned by the institution 有关机构拥有的资产类别的负面宣传
 - ✓ Increased potential for deterioration in the institution's financial condition 机构财务状况恶化的潜在增加
 - ✓ Widening debt or credit default swap spreads 债务或信用违约掉期利差扩大
 - ✓ Increased concerns over the funding of off-balance-sheet items 表外项目融资的增加

Chapter 4 The Investment Function in Financial-Services Management(☆☆☆)

1. Invest Instruments

【投资工具按照不同的标准有很多分类，我们这里主要按照时间分类】

● Money market instruments 货币市场工具

- ◇ 到期时间小于一年
- ◇ 风险低，收益低
- ◇ 容易交易

● Capital market instruments 资本市场工具

- ◇ 到期时间大于一年
- ◇ 收益相对高

● Key advantages and disadvantages of Money market Instruments

Type	Key Advantages	Key Disadvantages
Treasury Bills	Safety and high liquidity \Good collateral for borrowing\Can pledge behind government deposits	Low yields relative to other financial instruments\Taxable income
Short-Term Treasury Notes and Bonds	Safety, Good resale market\Good collateral for borrowing\Offer yields usually higher than T-bill yields	More price risk than T-bills\Taxable gains and income
Federal Agency Securities	Safety, Good to average resale market\Good collateral for borrowing\Higher yields than on U.S. government securities	Less marketable than Treasury securities\Taxable gains and income
Certificates of Deposit	Insured to at least \$100,000\Yields higher than T-bills\Large denominations often marketable through dealers	Limited resale market on longer-term CDs\Taxable income
International Eurocurrency Deposits	Low risk\Higher yields than on many domestic CDs	Volatile interest rates\Taxable income
Bankers' Acceptances	Low risk\Due to multiple credit guarantees	Limited availability at specific maturities\Issued in odd denominations\Taxable income
Commercial Paper	Low risk due to high quality of borrowers	Volatile market\Poor resale market\Taxable income
Short-Term Municipal	Tax-exempt interest income	Limited resale market\Taxable capital gains

Obligations		
✓ 美国国库券（Treasury Bills）：美国政府发行的短期债务工具，期限小于 1 年。		
✧ 具有高安全性，由联邦政府的征税权支持，市场价格相对稳定，且易于交易。		
✓ 短期国债（Short-Term Treasury Notes and Bonds）		
✧ 最初发行时具有较长到期时间的国债，当其到期时间在一年以内时被视为货币市场工具。		
✧ 比 Treasury Bills 对利率风险更敏感，市场流动性较低，但预期收益通常较高。		
✓ 联邦机构证券（Federal Agency Securities）		
✧ 由联邦政府或赞助的机构发行的可市场化债券。		
✧ 尽管大多数不受联邦政府正式担保，但市场普遍认为政府会在机构遇到困难时提供救助，从而保持其收益率接近国债，并维持高流动性。		
✓ 存单（Certificates of Deposit, CDs）		
✧ 存款机构发行的利息票据，为存款机构提供额外的资金来源。		
✧ 具有固定期限，小额面向消费者，大额面向企业和机构。具有固定或浮动利率。		
✓ 欧洲货币存款（Eurocurrency Deposits）		
✧ 全球大型银行发行的定期存款，通常在伦敦市场。		
✧ 期限短，收益率略高于国内存单。		
✓ 银行承兑汇票（Bankers' Acceptances）		
✧ 银行承诺在未来指定日期支付指定金额的票据，通常用于出口、进口或存储货物的信用担保。		
✧ 市场上最安全的货币市场工具之一，具有二级市场，可以在到期前转售。		
✓ 商业票据（Commercial Paper）		
✧ 大公司发行的短期无担保借据。		
✧ 短期到期，通常由信用评级很高的借款人发行。欧洲市场的票据到期时间较长，收益率较高，但信用风险较大。		
✓ 短期市政债务（Short-Term Municipal Obligations）		
✧ 州和地方政府发行的短期债务工具。		
✧ 利息收入免于联邦所得税，但信用质量受金融危机影响有所削弱。		

● Key advantages and disadvantages of Capital market instruments

Type	Key Advantages	Key Disadvantages
Treasury Notes and Bonds	Safety\Good resale market\Good collateral for borrowing\May be pledged behind government deposits	Low yields relative to long-term private securities\Taxable gains and income\Limited supply of longest-term issues
Municipal (State and Local Government) Bonds	Tax-exempt interest income\High credit quality\Liquidity and marketability of selected securities	Volatile market\Some issues have limited resale potential\Taxable capital gains
Corporate Notes and Bonds	Higher pretax yields than on government securities\Aid in locking in higher long-term rates of return	Limited resale market\Inflexible terms\Taxable gains and income
Asset-Backed Securities	Higher pretax yields than on Treasury securities\Collateral for borrowing additional funds	Less marketable and more unstable in price than Treasury securities\May carry substantial default risk\Taxable gains and income

✓ 国债（Treasury Notes and Bonds）

- ✧ T-notes: 原始到期时间（1 年至 10 年）和大量发行。
- ✧ T-bonds: 原始到期时间超过 10 年，市场交易较少，价格波动较大。
- ✧ 与国库券（T-bills）相比，T-notes 和 T-bonds 预期收益较高，但价格和流动性风险也更大。
- ✧ 一般面额有\$1,000、\$5,000、\$10,000、\$100,000 和\$1 百万。

✓ 市政票据和债券（Municipal Notes and Bonds）

- ✧ 由州、市和其他地方政府单位发行的长期债务工具，统称为市政债券。
- ✧ 大多数市政债券的利息免于美国联邦所得税，但资本利得全额征税。
- ✧ 市政债券流动性较差。

✓ 公司票据和债券（Corporate Notes and Bonds）

✓ Structured Notes、Securitized Assets

2. Factors affecting choice of investment securities

① Expected rate of return: YTM or HPY

- ✧ 到期收益率（YTM）：用于有到期日的证券。
- ✧ 持有期收益率（HPY）：用于无到期日或提前出售的证券。

② Tax exposure

- ✧ 当税率较高时，倾向购买免税证券
- ✧ 公司债券的税后收益率 $\text{After-tax gross yield} = \text{YTM} \times (1 - \text{税率})$
- 税率: marginal income tax rate

$$\begin{aligned} \text{tax - equivalent yield (TEY)} &= \frac{\text{市政债券收益率}}{1 - \text{税率}} \\ &= \frac{\text{After tax return on a tax - exempt investment}}{(1 - \text{firm's marginal tax rate})} \end{aligned}$$

- 市政债券是免税的
- TEY: 税收等效收益率
- ✧ 例子: 假设银行正在考虑购买 AA 级公司债券 (到期收益率 6%) 或 AA 级市政债券 (到期收益率 4.4%)。公司税率为 35%。税后收益率:
 - 公司 AA 债券: $6\% \times (1 - 0.35) = 3.9\%$
 - 市政 AA 债券: 4.4% (免税)
 - $3.9\% < 4.4\%$, 市政债券更有吸引力, 因为其税后收益率更高。
 - 税收等效收益率 (TEY): 比较免税和应税投资的税前收益率。示例中, $\text{TEY} = 4.4\% / (1 - 0.35) = 6.8\%$ 。
- ✧ Tax swap 税收互换
 - 是一种税务策略。在贷款收入较高的年份, 税务互换可能带来收益。
 - In a tax swap, the lending institution **sells lower-yielding securities at a loss** in order to reduce its current taxable income, while simultaneously **purchasing new higher-yielding securities** in order to boost future returns. 【其目的是通过出售低收益的证券来实现资本损失, 从而减少当期的应税收入, 同时再投资于更高收益的证券, 以增加未来的投资回报。】

③ Interest rate risk

- ✧ 利率变动不仅影响投资组合的价值, 还影响贷款需求。银行可以通过利率掉期等衍生品对冲风险。

④ Credit or default risk: including downgrade risk

⑤ Business risk: different business cycle

⑥ Liquidity risk

⑦ Call risk: holding callable instruments

⑧ Prepayment risk: whether prepayment or extension

⑨ Inflation risk: TIPS

⑩ Pledging requirements 质押要求: providing source of liquidity

3. Investment Maturity Strategies 投资期限策略

在投资组合的构建时，考虑期限的策略(不仅关注收益，还重点关注流动性)

(1) The Ladder or Spaced-Maturity Policy 梯形策略

- ✓ Strategy: Divide investment portfolio **equally among all maturities** acceptable to the investing institution. 将证券均匀分布在不同到期区间，降低收入波动。
- ✓ Advantages: Reduces investment income fluctuations/**requires little management expertise**. 执行简单

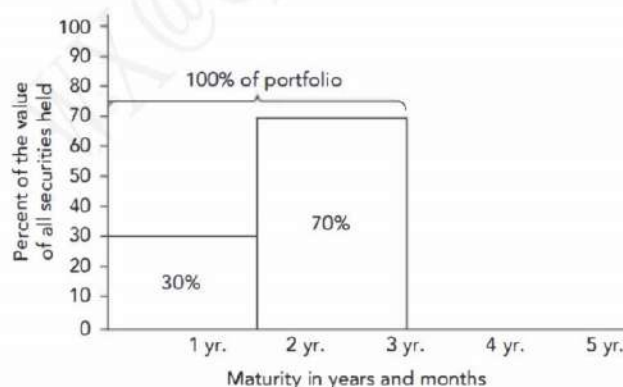
The Ladder or Spaced-Maturity Policy



(2) The Front-End Load Maturity Policy 前端集中策略

- ✓ Strategy: **All security investments are short-term**. 投资于短期证券
- ✓ Advantages: Strengthens the financial firm's liquidity position and avoids large capital losses if market interest rates rise. 主要关注流动性而非收入

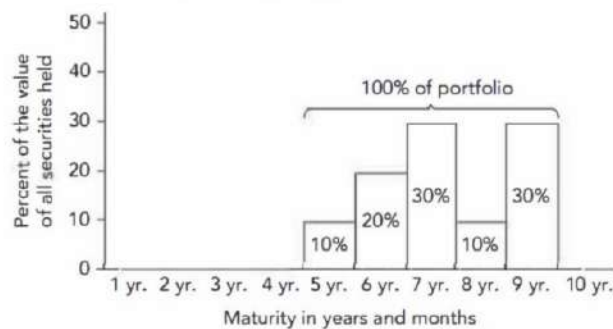
The Front-End Load Maturity Policy



(3) The Back-End Load Maturity Policy 流动性而非收入

- Strategy: **All security investments are long-term**. 投资于长期证券
- Advantages: Maximizes income potential from security investments if market interest rates fall. 主要关注收益。

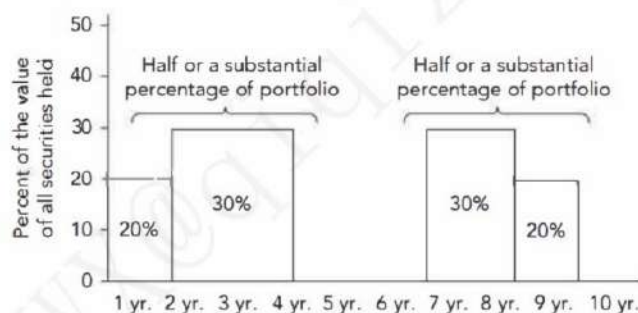
The Back-End Load Maturity Policy



(4) **The Barbell Strategy** 杠铃策略【这个知识点在一级学过】

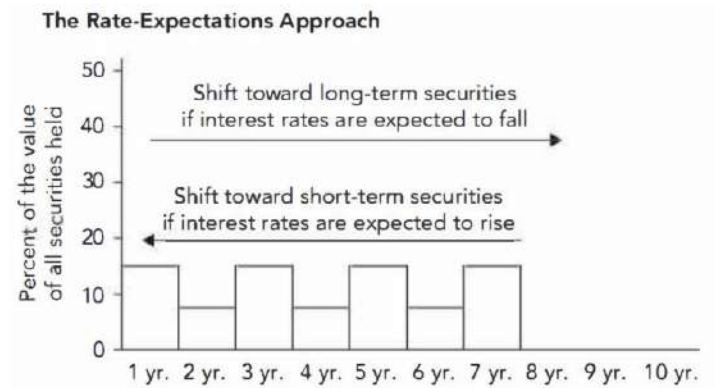
- ✓ Strategy: Security holdings are divided between **short-term and long term**. 结合短期和长期投资
- ✓ Advantages: Helps to meet liquidity needs with short term securities and to achieve earnings goals due to higher potential earnings from the long term portion of the portfolio. 同时考虑流动性和收益
- What is more, it **gains large convexity benefit**. 凸性越大, 杠铃策略越有利。

The Barbell Investment Portfolio Strategy



(5) **The Rate Expectations Approach** 利率预期策略【根据对未来利率的预测进行投资】

- ✓ Strategy: Change the **mix** of investment maturities as the interest-rate outlook changes. 根据利率和经济预测不断调整投资期限。
- ✓ Advantages: **Maximizes the potential for earnings** (and also for losses). 潜在收益高但风险和交易成本也大



● Five different investment maturity strategies

- ✓ No opinion no money → The Ladder, or Spaced-Maturity Policy
- ✓ Liquidity matters → The Front-End Load Maturity Policy
- ✓ Return matters → The Back-End Load Maturity Policy
- ✓ Both liquidity and return matter → The Barbell Strategy
- ✓ Money and ideas → The Rate Expectations Approach

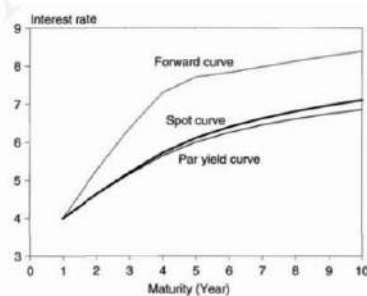
4. Maturity Management Tools 期限管理工具

在选择不同期限的投资时，需要运用两个重要的工具——收益率曲线和久期。

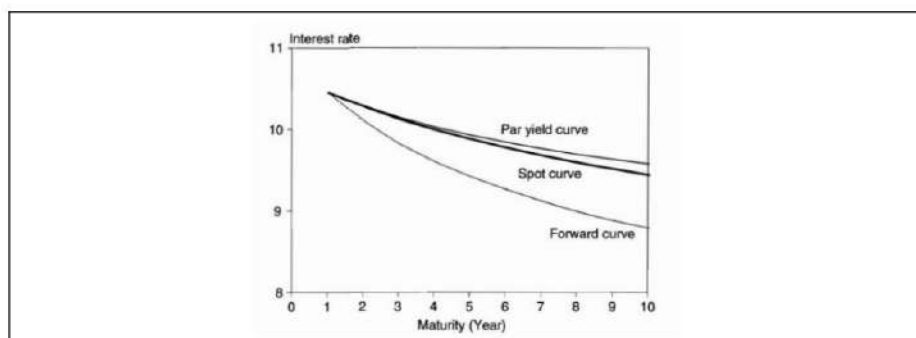
(1) The Yield Curve 收益率曲线：

收益率曲线相关知识回顾（一级知识）

- 利率期限结构(也就是收益率曲线)指的是利率和对应的期限之间的关系。
- 利率期限结构形状：
 - ✓ the term structure is **flat**: 长期利率=短期利率；
 - ✓ upward sloping 向上倾斜: 长期利率>短期利率



- ✓ downward sloping 向下倾斜: 长期利率<短期利率



- 通过分析收益率曲线，可以了解不同期限的利率趋势，从而预测未来利率变化，并相应调整投资组合。

- 赚钱的策略：

- ✓ **Pursuing the Carry Trade** 追逐利差交易

- ✧ **Pursuing the Carry Trade:** borrow at lower short-term rates to invest in higher-yielding longer-term securities. 【借短投长，短期利率低，长期利率高】

- 低利率的短期贷款（比如一个月的贷款）去买高利率的长期投资（比如五年的政府债券）。

- ✧ 隐含假设：利率期限结构向上倾斜

- ✧ The difference between these two rates of return is called **carry income** and tends to be **greatest** when the yield curve has a **steep upward slope**. 【收益率曲线向上倾斜，并且越陡峭，短期利率和长期利率的差距越大，因此这种策略所获收益越大。】

- ✓ **Riding the Yield Curve** 骑行收益率曲线 【关于这个策略，原版书介绍得很简略，不到 150 个字】

- ✧ 具体操作：

- 寻找一些即将到期的证券，这些证券的价格已经显著上升，而到期收益率下降。

- 卖出这些证券，赚取由于价格上涨带来的资本收益。

- 再投资：卖出证券后的资金，再投资于长期证券。

- ✧ 潜在假设：收益率曲线向上倾斜 **upward sloping**，并保持不变

(2) Duration

- 用 duration 构建免疫策略(Immunization).

- Portfolio immunization 免疫策略 is a strategy to protect (i.e., immunize) a portfolio's return from changes in interest rates, regardless of which direction interest rates move. 【免疫策略本质上就是久期对冲】

- ✓ 概念介绍：【关于概念介绍部分是基础内容，我这里重新调整了下】

- ◇ 当投资期限 investment horizons 不同时，投资者面临的利率风险是不同的。
 - Market price risk matters more when the investor has a short-term investment horizon. 【投资期限比较短时，Market price risk (或 price risk) 占主导地位】
 - Coupon reinvestment risk matters more when the investor has a long-term investment horizon. 【投资期限比较长时，reinvestment risk 占主导地位】
- ◇ investment horizon > Macaulay duration: 长期投资 → coupon reinvestment risk dominates market price risk → 利率下降，利率风险越大
- ◇ **investment horizon = Macaulay duration: coupon reinvestment risk offsets market price risk (利率风险被对冲，即免疫了) → → Immunization 免疫策略**
- ◇ investment horizon < Macaulay duration: 短期投资 → market price risk dominates coupon reinvestment risk → 利率上升，利率风险越大
- ✓ **Immunization 免疫策略【让投资组合的期限等于久期】**
 - ◇ **investment horizon = Macaulay duration**

Chapter 5 Liquidity and Reserves Management: Strategies and Policies(☆☆☆)

1. Net Liquidity and Strategies

(1) net liquidity position (L)净流动性

net liquidity position (L)净流动性= 流动性供给 supply for liquidity- 流动性需求 demand for liquidity

$$\begin{array}{l}
 \text{A financial firm's net liquidity position (L)} = \begin{array}{c} \text{Supplies of Liquidity Flowing into the Financial Firm} \\ \text{Incoming deposits (inflows)} + \text{Revenues from the sale of nondeposit services} + \text{Customer loan repayments} + \text{Sales of assets} + \text{Borrowings from the money market} \end{array} \\
 \quad \quad \quad - \begin{array}{c} \text{Demands on the Financial Firm for Liquidity} \\ \text{Deposit withdrawals (outflows)} - \text{Volume of acceptable loan requests} - \text{Repayments of borrowings} - \text{Other operating expenses} - \text{Dividend payments to stockholders} \end{array}
 \end{array}$$

- 流动性供给是指银行可以利用的可用资金
- 流动性需求是指银行需要满足的所有现金支出
- 流动性需求：包括
 - ✓ customers withdrawing money from their accounts 【客户把钱取走】
 - ✓ credit requests from customers the institution wishes to keep, either in the form of new loan requests or drawings upon existing credit lines 【客户借钱】
 - ✓ paying off previous borrowings, such as loans the institution may have received from other financial firms or from the central bank 【偿还以前的借款】
 - ✓ Operating expense 【银行日常运营费用】
 - ✓ payment of income taxes or cash dividends to stockholders periodically gives rise to a demand for immediately spendable cash 等【支付税款或现金股息等】
- 流动性供给：包括
 - ✓ receipt of new customer 【收到新客户的资金】
 - ✓ customers repaying their loans 【客户还偿还贷款】
 - ✓ sales of assets 【出售资产】
 - ✓ revenues (fee income) generated by selling nondeposit services 【销售非存款服务所产生的收入】

- ✓ borrowings in the money market 【在货币市场借款】
- 流动性需求具有重要的时间维度 time dimension，有些是即时的（比如第二天储户就要取走大额资金），有些是长期的。

(2) 金融机构的流动性管理问题：

- Rarely are demands for liquidity equal to the supply of liquidity at any particular moment in time. The financial firm must continually deal with either a liquidity deficit or a liquidity surplus.
 - ✓ 流动性需求等于流动性供给，这种情况在金融机构极少出现
 - ✓ 金融机构常常面临的情况是：流动性赤字或流动性盈余
- There is a trade-off between liquidity and profitability 流动性与盈利能力之间要做权衡。
 - ✓ 为了满足流动性需求，也就是流动性盈余，金融机构的预期盈利能力就越低（其他因素保持不变）。
 - ✓ 为了盈利，可能存在流动性赤字，很有可能面临流动性风险。

(3) 流动性管理策略：3 类

- Asset Liquidity Management (or Asset Conversion) Strategies 资产流动性策略
 - ✓ 也称 Asset conversion strategy
 - ✓ 将流动性存储^①在资产（主要是现金和流动性较高的资产）中【说白了就是在自己的账上，保留一些流动性较好的资产】
 - ✓ 当需要流动性时，将资产卖掉转换为现金。
 - 通常使用流动性较高的资产：如国库券、联邦基金贷款、市政债券等
 - ✓ 小金融机构比较适合，也是最古老的流动性管理的办法
 - ✓ 优劣势：比借款风险低，但面临机会成本、交易成本和资产价格可能下降的风险。
- Borrowed Liquidity (Liability) Management Strategies 负债流动性管理
 - ✓ relying on borrowed liquidity to meet cash demands.借钱以满足预期的流动性需求。
 - ✓ 也称：purchased liquidity or liability management
 - ✓ 借款来源：包括可转让的大额存单、联邦基金借款、回购协议、欧元货币借款以及从中央银行借款。
 - ✓ 大型金融机构常用

- ✓ 优缺点：
 - 银行可以仅在**需要时借款**，保持原有的**资产组合不变**，并根据需求**调整借款利率**。
 - 由于**利率波动和借款成本的不确定性**，这种方法具有较高的风险。
 - **Balanced Liquidity Management Strategies**
 - ✓ 资产管理和负债管理策略结合起来。一些流动性需求存储在资产中，而另一些流动性需求则通过借入资金来满足。
 - ✓ 通常：借钱满足短期意外的流动性需求。而长期的需求需要进行规划，通常将资金存储在短期和中期资产中，来应对这种流动性需求。
 - ✓ Most financial firms compromise by using both asset and liability management.
2. Estimating Liquidity Needs 估计流动性需求：4 种方法
- 【每种方法都基于特定的假设，并且只能给出对实际流动性需求的近似估计】
- the sources and uses of funds approach 资金来源与运用法
 - the structure of funds approach 资金结构法
 - the liquidity indicator approach 流动性指标法
 - the market signals (or discipline) approach 市场信号法【考纲未做要求，下面将不做介绍】
- (1) The sources and uses of funds approach 资金来源与运用法
- liquidity gap: difference between sources and uses of funds
 - ✓ **positive liquidity gap**, 也称 liquidity surplus 流动性盈余
 - sources of liquidity 资金来源 > 资金运用 uses of liquidity → positive liquidity gap (surplus)
 - 可能要考虑投资问题
 - ✓ **negative liquidity gap** , 也称 liquidity deficit 流动性赤字
 - 资金来源 < 资金运用时, → negative liquidity gap (deficit)
 - 可能要考虑融资问题。
 - **Estimated liquidity deficit or surplus = Δ deposits - Δ loans**
 - ✓ 当存款增加 (Δ deposits > 0) 而贷款减少 (Δ loans < 0) 时，流动性增加；
 - ✓ 当存款减少而贷款增加时，流动性减少。
 - 通常将存款和贷款变化的预测划分为三个组成部分：trend component 趋势、

seasonal component 季节性和 cyclical component 周期性。

● 原版书例子：【注意红色框框数据的关系】

- ✓ A bank whose trend growth rate in deposits and loan have averaged 10 percent and 8 percent a year respectively.
- ✓ Bank's total deposits stood at \$1,200 million and total loans outstanding reached \$800 million at the most recent year-end.

Deposit Forecast for	Trend Estimate for Deposits	Seasonal Element*	Cyclical Element**	Estimated Total Deposits
January, Week 1	\$1,210 +	-4	-6	= \$1,200
January, Week 2	1,212	-54	-58	1,100
January, Week 3	1,214	-121	-93	1,000
January, Week 4	1,216	-165	-101	950
February, Week 1	1,218	+70	-38	1,250
February, Week 2	1,220	+32	-52	1,200
Loan Forecast for	Trend Estimate for Loans	Seasonal Element*	Cyclical Element**	Estimated Total Loans
January, Week 1	\$799	+6	-5	\$800
January, Week 2	800	+59	-9	850
January, Week 3	801	+174	-25	950
January, Week 4	802	+166	+32	1,000
February, Week 1	803	+27	-80	750
February, Week 2	804	+98	-2	900

Time Period	Estimated Total Deposits	Estimated Total Loans	Estimated Deposit Change	Estimated Loan Change	Estimated Liquidity Deficit (-) or Surplus (+)
January, Week 1	\$1,200	\$ 800	\$ ____	\$ ____	\$ ____
January, Week 2	1,100	850	-100	+50	-150
January, Week 3	1,000	950	-100	+100	-200
January, Week 4	950	1,000	-50	+50	-100
February, Week 1	1,250	750	+300	-250	+550
February, Week 2	1,200	900	-50	+150	-200

- ✓ 在这个例子中，需要做哪些流动性管理决策？
- ✓ 要考虑在第 2、3、4 和 6 周→筹集新资金（流动性赤字）
 - 具体比如：关注流动性资金来源，首先评估银行的流动性资产（看看哪些资产可用），然后确定是否能够借入资金等。
- ✓ 第 5 周盈余→考虑投资（流动性盈余）

(2) The structure of funds approach 资金结构法

- 这种方法的核心：将存款和其他资金，根据其提款的可能性进行分类，然后相应地设置流动性准备金 total liquidity requirement。
- 资金分类：根据其提款可能性
 - ✓ “热钱” 负债 Hot money" liabilities
 - 也称 volatile liabilities。利率高度敏感的资金，预计在当前期间就会

被提取。

- ✓ 易受损资金 **Vulnerable funds-customer deposits**: 可能有相当一部分资金在当期被提取。介于 **Hot money** 和 **Stable funds** 之间
- ✓ 稳定资金 **Stable funds**
 - 也称 **core deposits** 核心存款 or **core liabilities** 核心负债。不太可能在当期提取的资金
- 设置流动性准备金 **total liquidity requirement**:
 - ✓ **Liability liquidity reserve** 负债流动性需求 $= 0.95 \times (\text{Hot money deposits and nondeposit funds} - \text{Required legal reserve}) + 0.30 \times (\text{Vulnerable deposit and nondeposit funds} - \text{Required legal reserve}) + 0.15 \times (\text{Stable deposits and nondeposit funds} - \text{Required legal reserve})$
 - Required legal reserves is usually 3% of the total amount
 - ✓ **loan liquidity requirement** 贷款流动性需求 $= 1.00 \times (\text{Potential loans outstanding} - \text{Actual loans outstanding})$
 - 贷款流动性需求: 机构的目标是批准所有优质贷款, 所以前面的系数是 1。
 - ✓ **Total liquidity requirement** 总流动性需求 = 负债流动性需求 + 贷款流动性需求
- 原版书例子:
 - ✓ First National Bank finds that its current deposits and nondeposit liabilities break down as follows:
 - Hot money → \$ 25 million
 - Vulnerable funds → \$ 24 million
 - Stable (core) funds → \$100 million
 - ✓ First National Bank's management wants to keep a 95% reserve behind its hot money deposits (less the 3% legal reserve requirement imposed by the central bank behind many of these deposits) and nondeposit liabilities, a 30% liquidity reserve in back of its vulnerable deposits and other borrowings (less required reserves), and a 15% liquidity reserve behind its core deposit and nondeposit funds (less required reserves).
 - ✓ First National Bank's loans total \$135 million but recently have been as high as \$140 million, with a trend growth rate of about 10 percent a year. This financial firm wishes to be ready at all times to honor customer demands for all loans that meet its quality standards.

- ✓ Calculate the total liquidity requirement of First National Bank.

解答:

bank's total liquidity requirement = $0.95 (\$25 \text{ million} - 0.03 \times \$25 \text{ million})$
 $+ 0.30 (\$24 \text{ million} - 0.03 \times \$24 \text{ million}) + 0.15 (\$100 \text{ million} - 0.03 \times \$100 \text{ million})$
 $+ \$140 \text{ million} \times 0.10 + (\$140 - \$135 \text{ million})$
 $= \$63.57 \text{ million (held in liquid assets and additional borrowing capacity)}$

- 概率分析: 有些机构使用概率来评估预期的流动性需求。
- ✓ 定义不同的流动性情景, 并为每种结果分配概率。
 - Best possible liquidity position 最好的情况
 - Liquidity position bearing the highest probability 最高概率的情况
 - Worst possible liquidity position 最差的情况
- ✓ Expected liquidity requirement = Probability of Outcome A $\times$ (Estimated liquidity surplus or deficit in Outcome A) + Probability of Outcome B $\times$ (Estimated liquidity surplus or deficit in Outcome B) +
- ✓ 管理层还应制定应对不利情景的应急计划
- ✓ 原版书例子:

Possible Liquidity Outcomes for Next Week	Estimated Average Volume of Deposits Next Week (millions)	Estimated Average Volume of Loans Next Week (millions)	Estimated Liquidity Surplus or Deficit Position Next Week (millions)	Probability Assigned by Management to Each Possible Outcome
Best possible liquidity position (maximum deposits, minimum loans)	\$170	\$110	+\$60	15%
Liquidity position bearing the highest probability	\$150	\$140	+\$10	60%
Worst possible liquidity position (minimum deposits, maximum loans)	\$130	\$150	-\$20	25%

- ✓ Expected liquidity requirement = $0.15 \times (+\$60 \text{ million}) + 0.60 \times (+\$10 \text{ million}) + 0.25 \times (-\$20 \text{ million}) = +\$10 \text{ million}$

(3) The liquidity indicator approach 流动性指标法

【许多金融服务机构根据经验和行业平均水平来估计其流动性需求】

- 下列比率指标越高, 流动性越好
 - ✓ Cash position indicator: (cash + deposits due from depository

- institutions) / total assets.
- ✓ Liquid securities indicator 流动证券指标: U.S. government securities / total assets.
 - 持有政府证券的比例越高, 存款机构的流动性通常越好。
 - ✓ Core deposit ratio 核心存款比率: Core deposits / total assets.
 - Core deposits 核心存款包括少于 10 万美元的全部存款。核心存款主要是不太可能在短时间内提取的小额支票和储蓄账户, 因此具有较低的流动性要求。
 - ✓ Hot money ratio 热钱比率: Money market (short-term) assets / volatile liabilities.
 - Money market assets are cash, monies due from deposits at other institutions, short-term securities holdings, reverse repo agreements, and federal funds loans.
 - Volatile liabilities are large CDs, Eurocurrency deposits, repo agreements, and federal funds borrowings.
 - ✓ Net federal funds and repurchase agreements position 净联邦基金和回购协议头寸: (Federal funds sold and reverse repo agreements - federal funds purchased and repo agreements) / total assets.
 - 下列指标比率越高, 流动性越不好:
 - ✓ Capacity ratio 容量比率: Net loans and leases / total assets.
 - 贷款和租赁往往是最不流动的资产
 - ✓ Loan commitments ratio: Unused loan commitments / total assets.
 - Unused loan commitments represent promises of credit made by lenders to customers up a specific amount over a specific period of time.
 - ✓ Pledged securities ratio: Pledged securities / total securities holdings.
 - 因为抵押率越高, 可供出售的证券就越少。
 - ✓ Deposit composition ratio: Demand deposits / time deposits.
 - Demand deposits fluctuate significantly while time deposits have fixed maturities and typically assess penalties for early withdrawal.
 - ✓ Deposit brokerage index: Brokered deposits / total deposits.
 - Brokered deposits are funds placed by securities brokers for their customers which are interest sensitive and may be withdrawn quickly.

3. Legal Reserve 法定准备金

- **法定准备金 Legal Reserve**: 法律和央行规定在特定时间段内 **必须持有** 的资产。

小背景:

- ✓ 在美国, 合格的存款机构(qualified depository institution)必须以现金形式持有所需的法定准备金, 如果不够, 还必须以存款形式存放在该地区的联邦储备银行(Federal Reserve bank)的储备账户(reserve account)中。
- ✓ 较小的存款机构和银行, 如果不是联邦储备系统(Federal Reserve system)的成员, 其法定准备金存款可以放在经联邦储备银行批准的机构中。
- ✓ 受美国法定准备金要求约束的金融机构: 商业银行、储蓄银行、储蓄贷款协会、信用合作社、在美国提供合格存款和其他负债的外国银行的代理机构和分支机构。

● 准备金计算

	不是当前的金额, 是过去的金额 → 滞后准备金核算 lagged reserve accounting (LRA)		每年都会变(2010年): ✧ 对于交易存款(例如支票账户等), 10.7million-58.8million→3% ✧ 如果同一存款机构持有超过 58.8million 的交易存款, 超过部分→10% ✧ 非交易性负债(包括非个人定期存款和欧元货币负债)→准备金为0%
法定准备金=	金额	*	准备金率
total required legal reserves = (transaction deposit reserve requirement × daily average amount of net transaction deposits for the computation period) + (nontransaction reservable liabilities' reserve requirement × daily average amount of nontransaction deposits for the computation period)			

● 原版书例子

- ✓ First National's net transaction deposits averaged \$100 million over the 14-day reserve computation period, while its nontransaction reservable liabilities had a daily average of \$200 million over the same period.
- ✓ First National held a daily average of \$5 million in vault cash over the required two-week reserve computation period.

解答:

- ✓ average required legal reserve level = $0.0 \times \$10.7 \text{ million} + 0.03 \times (\$58.8 \text{ million} - \$10.7 \text{ million}) + 0.10 \times (\$100 \text{ million} - \$58.8 \text{ million}) = \5.563 million .
- ✓ Daily average level of additional = Total required - Daily average vault =

\$5.563 million - \$5.000 million = \$0.563 million

✓ Tips: First National 必须再补充 0.563 million 才能达到法定存款准备金要求

● sources of reserves

【当准备金不足时，需要补充资金，以达到监管要求。】

✓ Immediacy of need 需求的紧急性：

- 特别紧急：通过隔夜贷款市场或向中央银行的贴现窗口借款来解决，
- 非紧急情况：出售存款或资产。

✓ Duration of need 需求的持续时间：如果流动性赤字预计只会持续几小时，那么隔夜贷款市场或中央银行的贴现窗口通常是首选。然而，持续几天、几周或几个月的流动性短缺通常会通过出售较长期的资产或较长期的借款来解决。

✓ Access to the market for liquid funds 获取流动性资金的途径

- 例如，小型机构可能无法在欧洲外汇市场（Eurocurrency market）借款或发行商业票据。

✓ Relative costs and risks of alternative sources of funds 替代资金来源的相对成本和风险：

- 选择成本最低、可靠性最高的资金来源。

✓ The interest rate outlook 利率展望：选择未来利率预计最低的资金来源。

✓ Outlook for central bank monetary policy 中央银行货币政策展望。

✓ Rules and regulations applicable to a liquidity source. 适用于流动性来源的规则和法规

- 大多数流动性来源不能随意使用，必须遵守规定。例如，从中央银行借款通常要求借款机构提供贷款背后的抵押品。在美国和欧洲，两个关键的流动性来源——联邦基金和欧元市场——借款机构需要交易日结束之前安排资金需求。

Chapter 6 Intraday Liquidity Risk Management(☆☆)

- 日间流动性 intraday liquidity 就是银行在一天中确保有足够的“零钱”来完成所有交易和支付，不出现资金短缺的问题。
- 管理日间流动性不仅是为了确保银行日常运营的顺利进行，还关系到整个金融系统的稳定和银行自身的风险管理与发展。因此，日间流动性管理是银行风险管理框架中的重要组成部分。

1. Uses and sources of intraday liquidity

● Use of intraday liquidity 日间流动性的使用

- ✓ 银行自己使用情况比较好预测，而客户使用情况相对不好预测

Funding Requirement (资金需求)	Description	Impacted by Client Activity (受客户活动影响)	Impacted by Bank Activity (受银行活动影响)	Ability to Forecast
Outgoing Wire Transfers (外发电汇) 【最大的日间流动性使用】	Payments on LVPS such as Fedwire and CHIPS (在大额支付系统如 Fedwire 和 CHIPS 上的支付)	Yes, clients provide bank instructions (是的，银行根据客户指示行动)	Yes, Bank Treasury and lines of business have payment needs (是的，银行司库和业务部门有支付需求)	Bank activity can usually be forecasted with 1 to 2 days' notice; client payment activity is more difficult to predict (银行活动通常可以提前 1 至 2 天预测；客户支付活动更难预测)
Settlements at Payment, Clearing and Settlement (PCS) Systems (支付、清算和结算系统的结算)	Net cash settlements at payments systems, clearinghouses (支付系统、清算所的净现金结算)	Not directly initiated by clients, but includes client activity (不是由客户直接发起，但包括客户活动)	Yes, position monitored by operations groups (是的，头寸由运营组监控)	Can be forecast for securities that have multi-day settlements (e.g., T+3); more difficult for same day settlement activity (可以预测多天结算的证券（例如 T+3）；同日结算活动更难预测)
Funding of Nostro Accounts (往来账户的资金)	Cash transfer to a correspondent bank for services provided (向代理银行转账)	Not directly initiated by clients, but includes client activity (不是由客户直接发起，但包括客户活动)	Yes, position monitored by operations groups (是的，头寸由运营组监控)	Securities settlements are generally predictable; client payment activity flowing through correspondents is less predictable (证券结算通常是可预测的；通过代理银行的客户支付活动较难预测)

Collateral Pledging (抵押品质押)	Obtaining and earmarking of collateral required by an outside beneficiary	Yes, some clients require collateral to cover bank trading liabilities or deposits (是的, 一些客户需要提交抵押品以覆盖银行交易负债或存款)	Yes, banks are required to post collateral for margin at FMUs or other trading counter-Parties (是的, 银行需要在金融市场基础设施或其他交易对手方处为保证金提供抵押品)	Dependent on trading volumes and asset price changes, generally known one day in advance (取决于交易量和资产价格变动, 通常提前一天知晓)
Asset Purchases/Funding (资产购买/融资)	Exchange of bank cash for another asset such as a client loan (将银行现金交换为另一种资产, 如客户贷款)	Yes, clients can draw down on lines of credit or letters of credit (是的, 客户可以动用信贷额度或信用证)	Yes, assets may be securities for the bank's investment portfolio or fixed assets (是的, 资产可能是银行投资组合中的证券或固定资产)	Bank fixed asset activity should be known in advance; securities purchases may be same day settlement; client loans are more difficult to predict (银行固定资产活动应提前知晓; 证券购买可能是同日结算; 客户贷款更难预测)

● Sources of intraday liquidity

Funding Requirement (资金需求)	Description	Client Activity (客户活动)	Bank Activity (银行活动)	Ability to Forecast (预测能力)
Cash Balances (现金余额)	Deposits at the central bank and at correspondent bank nostro accounts (在中央银行和代理银行往来账户的存款)	No, clients do not directly impact closing/start of day cash balances (否, 客户不会直接影响每天的开盘/收盘现金余额)	Yes, Bank Treasury determines level of closing/start of day cash balances (是的, 银行司库部决定每天的开盘/收盘现金余额水平)	Bank activity can usually be forecasted with 1 to 2 days' notice; client payment activity is more difficult to predict (银行活动通常可以提前 1 至 2 天预测; 客户支付活动更难预测)

Incoming Funds Flow (资金流入)	Incoming cash payments and cash credits from FMU (来自金融市场基础设施的现金支付和现金存款) financial market utilities (FMUs): 金融市场基础设施	Yes, incoming client payments are credited to the bank's accounts at the central bank and correspondents (是的, 客户的进账会记入银行在中央银行和代理银行的账户)	Yes, activities conducted for the bank's business can impact cash balances (是的, 为银行业务进行的活动可能会影响现金余额)	Bank activity can usually be forecasted with 1 to 2 days' notice; client payment activity is more difficult to predict. FMU credit can be forecast for securities that have multi-day settlements (e.g., T+3); more difficult for same day settlement activity (银行活动通常可以提前 1 至 2 天预测; 客户支付活动更难预测。对多天结算的证券 (例如 T+3) 的 FMU 信用可以预测; 同日结算活动更难预测)
Intraday Credit (日内信贷) 【其他机构给银行的额度】	Credit line or overdraft permitted during business hours and covered by close of business. Lines are often uncommitted and provided without interest charges (在营业时间内允许的信贷额度或透支。这些额度通常是非承诺的, 并且不收取利息)	No, client activity does not directly impact the amount of intraday credit extended to a bank (否, 客户活动不会直接影响银行的日内信贷额度)	Yes, counterparties will often adjust intraday credit extensions to reflect business activity from other areas of the bank (e.g., OTC trading) (是的, 对手方通常会调整日内信贷扩展, 以反映银行其他领域的业务活动 (例如场外交易))	Some intraday credit facilities are disclosed and well known to a bank (e.g., FRB net debit cap). Others are not disclosed but can often be inferred from historical data (一些日内信贷设施是公开的并为银行所熟知。其他的虽然未公开但通常可以从历史数据中推测出来)

Liquid Assets (流动资产)	Cash, money market deposits, and short-term government debt (e.g., T-Bills) which can be quickly converted to cash (现金、货币市场存款和短期政府债务(如国库券), 可以迅速转换为现金)	Yes, client may be sources of liquidity in converting liquid assets to cash (e.g., repo transaction) (是的, 客户可能是将流动资产转换为现金的流动性来源(例如回购交易))	Yes, bank trading and investment portfolio activity can impact the amount of liquid assets available (是的, 银行交易和投资组合活动可能会影响可用流动资产的数量)	Liquid assets held in the investment portfolio and in other money market investments tend to have low volatility and as a result are predictable (投资组合中持有的流动资产和其他货币市场投资往往波动性较低, 因此可预测)
Overnight Borrowings (隔夜借款)	Fed Funds, Eurodollar borrowing, overnight deposits (联邦基金、欧洲美元借款、隔夜存款)	Yes, clients may be direct sources of liquidity for overnight borrowing (e.g., overnight deposit) (是的, 客户可能是隔夜借款的直接流动性来源)	No, other Bank lines of business do not regularly supply overnight funding (否, 银行的其他业务部门不会定期提供隔夜资金)	Client supply of overnight borrowing tends to be fairly predictable, but with moderate volatility (客户提供的隔夜借款往往可预测, 但具有一定波动性)
Other Term Funding (其他定期资金)	Other term deposits, repos from FHLB, money market funds, etc. (其他定期存款、联邦住房贷款银行的回购协议、货币市场基金等)	Yes, clients may be direct sources of liquidity for term funding (是的, 客户可能是定期资金的直接流动性来源)	No, other Bank lines of business do not regularly supply term funding (否, 银行的其他业务部门不会定期提供定期资金)	Client supply of term funding tends to be fairly predictable, with low volatility (客户提供的定期资金往往可预测, 且波动性较低)

● The bank treasurer's challenge

【银行司库的主要任务是处理和管理来自银行各个部门的大量现金交易。这些交易数量庞大且来源复杂。司库需要确保银行在**日间使用的信贷额度不超标**, 同时还要保证在**每天结束时**, 银行账户里的**现金余额达到预定的目标**。也就是司库需要让银行每天保持资金的进出平衡, 并确保银行的资金状况稳定】

- ✓ the **variability** of cash flow patterns 现金流模式的可变性
 - 日常现金流的波动性高, 大型现金需求或来源的提前通知少。
- ✓ the impact of market forces 市场力量的影响
 - Daily volatility in asset prices can result in unanticipated margin calls that require additional cash funding. 资产价格的日常波动可能需要追加保证金, 需额外的资金。

- Central bank money desks that implement monetary policy directly may influence available liquidity in the marketplace, impacting a bank's ability to source or deploy intraday funds 中央银行的货币政策操作影响市场流动性。
- ✓ lack of real-time data 缺乏实时数据
 - Most bank treasurers today do not have a comprehensive, single source for real-time balances and expected transaction flows, which complicates the task of determining and forecasting cash throughout the day. 大多数银行司库缺乏综合的实时余额和预期交易流数据，增加了现金管理的复杂性。
- 2. Governance
 - ✓ **Active risk management**
 - 日间流动性风险被视为业务成本，结算风险和系统性风险(settlement and systemic risks) 视为其风险分类的一部分，将日间流动性风险纳入其风险管理框架，并积极管理
 - ✓ **Risk assessment:** as a component of risk self-assessments 自评估
 - ✓ **Integration with risk governance** 日间流动性风险管理的监督整合到银行的整体风险监督结构中
 - **Three lines of defense**, 分别是: **Treasury** (第一道防线), **risk management** (第二道防线) and **internal audit** (第三道防线)
 - ✓ **Risk measurement and monitoring:**
 - the amount of intraday credit the institution is extending to clients
 - ◆ 指的是在一个营业日内，银行向客户提供的短期贷款或透支额度
 - the amount of intraday credit the institution utilizes
 - ◆ 银行自身在一个营业日内使用的短期贷款或透支额度
- 3. Measurement of Intraday Liquidity
 - 【追踪银行日间流动性风险的常用衡量指标】
 - A. **Measures for Understanding Intraday Flows** 【单位是\$, 这一组指标有助于理解机构的日间流动性流动情况。】
 - **Total payments** 总支付额 【最好、最全】
 - ✓ maintain statistics concerning the amount of payments it makes on all electronic payments systems in which it participates. 【银行应统计所有电子支付系统的支付数据，包含金额、时间、处理步骤等信息】
 - **Other cash transactions**

- ✓ track its intraday and **end-of-day settlement** positions at all **FMUs** in which it participates, including **collateral** position in SSN. 【Securities settlements networks : SSN】
- **Settlement positions**
 - ✓ If complete data to reconstruct account positions **at any time of day is not available**, at a minimum a bank should maintain data on its settlement positions with all its FMUs. 至少应保留其在所有 FMU 中的结算头寸数据。
- Time sensitive obligations
 - ✓ Similar to settlement positions, these transactions require completion at **a specific time** during the day. 在特定时间完成的交易，例如市场活动、保证金支付等
- Total Intraday **Credit Lines to Clients and Counterparties** 客户使用的信贷额度
- Total Bank Intraday **Credit Lines Available and Usage** 其他机构给银行的可用和使用的银行日间信贷额度总额
- B. **Measures for Quantifying and Monitoring Risk Levels**(%是单位，这一组指标提供了用于监控风险水平的比率。)
- Daily Maximum Intraday Liquidity Usage
 - ✓ The measure is the ratio of **the day's largest net negative balance** relative to the size of **the committed or uncommitted credit line**.
- Intraday Credit Relative to Tier 1 Capital 【客户用到的额度/Tier 1 Capital】
- Client Intraday Credit Usage 【单个客户的透支额度/所有客户的信贷额度】
 - ✓ This measure is derived by comparing a client's peak daily intraday overdraft to the established (committed or uncommitted) credit line.
- Payment Throughput 支付吞吐量
 - ✓ These measures track the percentage of outgoing payment activity relative to time of day.

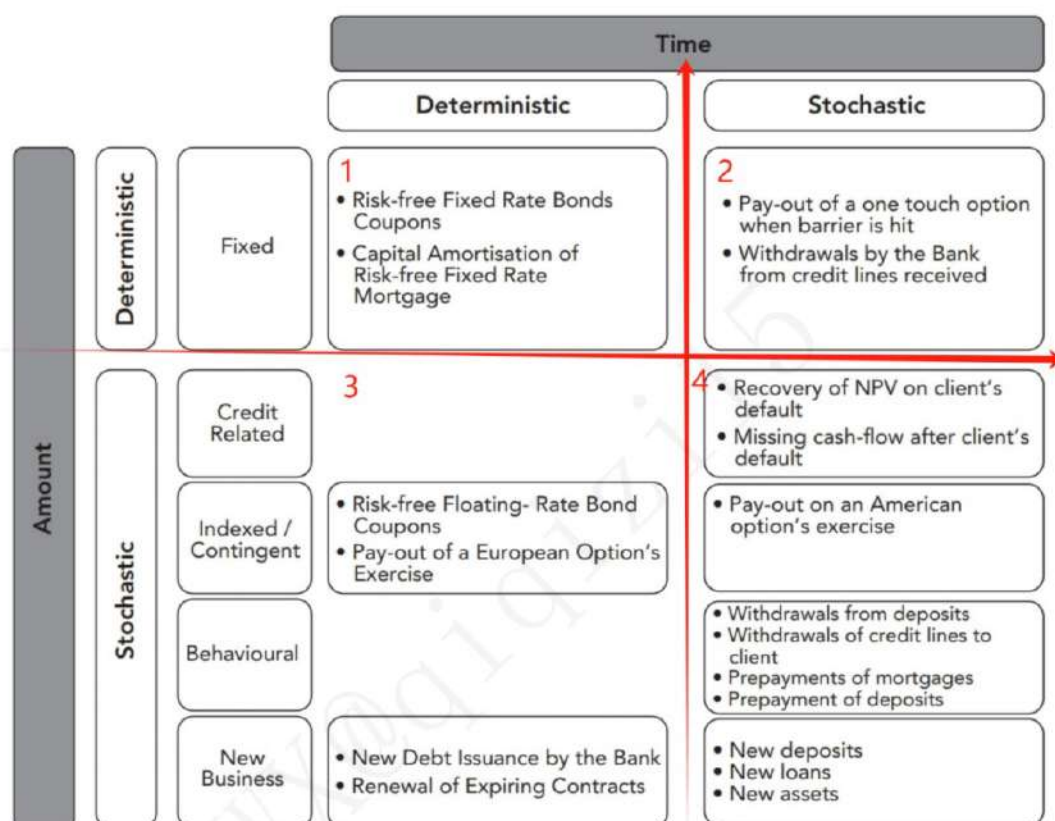
Chapter 7 Monitoring Liquidity(☆☆☆)

这一节内容比较多，也比较重要

1. Taxonomy of the cash flows 现金流的分类

● 现金流的分类维度：

- ✓ 时间维度：确定性 Deterministic 时间和随机 Stochastic 时间。
- ✓ 金额维度：确定性 Deterministic 金额和随机性 Stochastic 金额。



2. Liquidity Option 流动性期权

- ✓ **Liquidity Option 流动性期权**: a liquidity option can be defined as **the right of a holder** to receive cash from, or to give cash to, the bank at predefined times and terms. 【流动性期权，指的是权利持有人可以在未来的某个时间收到或支付现金的权利】
- ✓ 流动性期权的行权，会影响银行的现金流
- ✓ 流动性期权和普通期权的差别
 - ✧ 普通期权 Standard options are exercised when there is a profit. 【对于普通期权而言，当有 profit 时会行权】
 - ✧ Liquidity option is exercised because of the **cash flows produced after exercise, even if it makes loss**. 【对于流动性期权而言，即使有 loss 也可能行权。因为流动性期权更关注流动性而非是否盈利】

- ✓ 流动性期权使用的例子：
 - ✧ customer withdraw its unused credit line even if the credit spread shrinks and it would be cheaper for the obligor to get new funds in the market with a new loan 【unused credit line 信贷额度】
 - 银行为客户提供承诺的信用额度，客户有权在需要时提取资金，无论是由于信用利差的变化还是其他非财务原因。
 - ✧ the bank's clients can typically **withdraw all or part** of the deposited amount with no or short notice 客户可以随时提取全部或部分存款
 - ✧ client has the right to repay the funds before the contract ends, usually under low interest rate environment. 客户有权提前还款，这可能会对银行造成经济损失，但会增加银行的流动性
- 行使流动性期权对银行的影响
 - ✓ A liquidity impact on the balance sheet, given by the amount withdrawn or repaid.
 - 取款或偿还的金额会直接影响银行的流动性资产或流动性负债。例如，当客户从银行取款时，银行的流动性资产（如现金）会减少；反之，当银行偿还债务时，其流动性负债会减少。
 - ✓ A (positive or negative) financial impact, given by the difference between the contract's interest rates and credit spread and the market level of the same variables at the time the liquidity option is exercised, applied on the withdrawn or repaid amount.
 - 在行使流动性期权（如提前取款或提前偿还债务）时，可能会产生的财务影响。这些影响可以是正面的也可以是负面的，取决于以下几个因素：
 - ◆ 合同利率和信用利差 credit spread: 这是指在合同中规定的利率和信用利差。
 - ◆ 市场同类变量水平 the market level of the same variables: 这是指当前市场上同类产品或服务的利率和信用利差水平。
 - ◆ 合同利率和信用利差与市场同类变量水平之间的差异。
 - 例如，如果合同利率低于市场利率，那么提前取款可能会导致损失，因为公司可能需要支付更高的市场利率进行再融资。

3. The new definition of Liquidity risk

- **Funding cost risk:** The event that in the future the bank has to **pay greater than expected cost (spread)** above the risk-free rate to receive funds from

sources of liquidity that are available. 【融资时，融资成本偏离预期成本的风险】

● The **new definition of Liquidity risk**

- ✓ The amount of economic losses due to the fact that on a given date the algebraic sum of positive and negative cash flows and of existing cash available at that date, is **different from some (desired) expected level**.

【流动性风险：在某个特定日期，银行的所有收入和支出（正负现金流量），再加上当前持有的现金，合计起来的金额与预期的目标金额不一致而导致的经济损失。】

- ✓ 【考试时，如果没有明确说流动性风险的新定义，我们还是采用 chapter1 关于流动性风险的定义】

- ✓ 表现形式：

- ✧ ① Inability to raise enough funds to meet payment obligations, 【无法筹集足够的资金满足支付义务→融资流动性风险】
- ✧ ② Ability to raise funds only at costs above those expected. 【只能以高于预期的成本筹集资金】
- ✧ ③ Ability to invest excess liquidity only at rates below those expected 【只能以低于预期的利率投资多余的流动性】

4. **Quantitative Liquidity risk Measures**

- 银行在日常的经营过程中，会产生正、负现金流；现金流存在波动（更多关注负的现金流），需要银行司库部门进行管理（能产生正的现金流）。

✓ **Causes of liquidity 流动性成因→TSECF 和 TSECCF**

- ✧ All factors referring to existing and forecast future contracts originated by the **ordinary business activity** of a financial institution can be considered as the causes of liquidity risk.
- ✧ Cash flows generated by the causes of liquidity **can be both positive and negative**.
- ✧ 这种现金流是外生的，非司库所能决定，比如银行的投资部门，可能产生正的现金流，也可能产生负的现金流

✓ **Sources of liquidity 流动性来源→LGC 和 TSCLGC**

- ✧ All factors capable of generating **positive cash flows** to manage and hedge liquidity risk and can be disposed of promptly by the bank determine the **liquidity generation capacity(LGC)** of the financial

institution.能够迅速产生正现金流来管理和对冲流动性风险，决定了金融机构的流动性生成能力。

◇ **liquidity generation capacity(LGC)**流动性生成能力：获得正现金流的数量

◇ 这种现金流是内生的，司库可以自己决定的。

1) TSECF&TSECCF

● **The term structure of expected cash flows (TSECF)** 预期现金流的期限结构:未来特定日期的现金流，可正可负

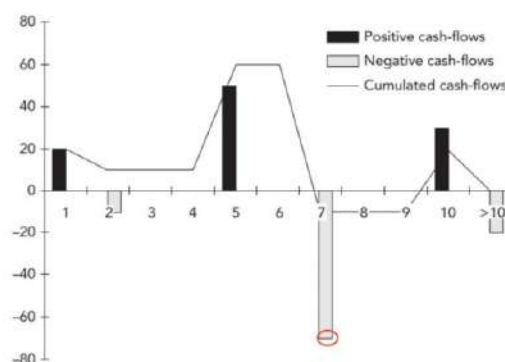
■ TSECF 涵盖所有现有合同和负债的现金流，这些现金流可能是随机的，因为它们可能与市场指数（如 Libor 或 Euribor ）相关。

● **The term structure of cumulated expected cash flows (TSECCF)** 累计预期现金流的期限结构:特定日期的累计现金流，理想状态下始终为正。

● The quantities entering in the TSECF and hence the TSECCF are **expected values**, since they all depend on **stochastic variables**.TSECF 和 TSECCF 期望值，实际情况中，现金流的时间和数量通常是随机 **stochastic** 的。

● 原版书例子：1.一个机构的简单资产负债表及对应的图形表示【只考虑本金】

Expiry	Assets	Liabilities
1	20	
2		10
5	50	
7		70
10	30	
>10		20 equity
	100	100



2.既考虑本金又考虑利息。具体利率如下表：

Expiry	Interest Yield	
	Assets (%)	Liabilities (%)
1	5.00%	
2		4.00%
5	6.00%	
7		4.50%
10	6.50%	
>10		

✓ 现金流展示: **Cf+** 现金流流入; **Cf-** 现金流流出

	Cf+		Cf-		TSECCF
	Notional	Interest	Notional	Interest	
1	20	5.95	0	-3.55	22.4
2	0	4.95	-10	-3.55	13.8
3	0	4.95	0	-3.15	15.6
4	0	4.95	0	-3.15	17.4
5	50	4.95	0	-3.15	69.2
6	0	1.95	0	-3.15	68
7	0	1.95	-70	-3.15	-3.2
8	0	1.95	0	0	-1.51
9	0	1.95	0	0	0.18
10	30	1.95	0	0	31.87
>10	0	0	-20	0	11.87

◇ 其中: $5.95 = 20 \times 5\% + 50 \times 6\% + 30 \times 6.5\%$

◇ $4.95 = 50 \times 6\% + 30 \times 6.5\%$

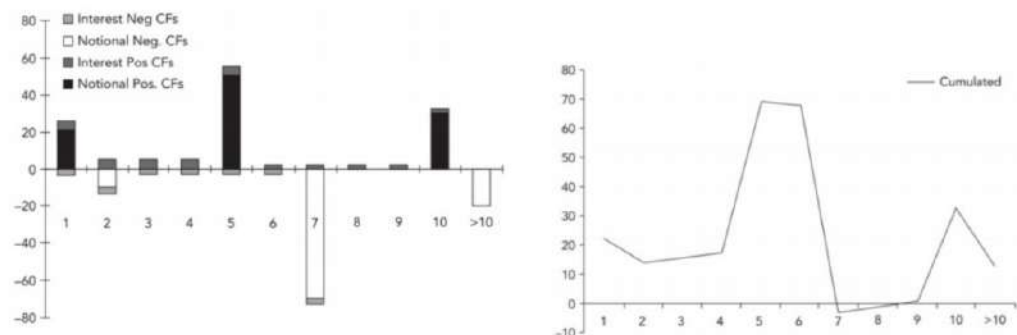
◇ $3.55 = 10 \times 4\% + 70 \times 4.5\%$

◇ $22.4 = 5.95 - 3.55 + 20$

◇ $13.8 = 22.4 + 4.95 - 3.55$

◇ Equity 的 20 元, 不会产生利息

◇ 表格中的数据依此类推。



● 包含在 TSECF 中的现金流:

【有时 TSECF 并不像上面这个例子这么简单、确定，现实中往往还需要考虑下面这些因素】

- ✓ ①考虑现有合同现金流的随机性：includes the cash flows from **all existing contracts** that comprise the assets and liabilities and remember in many cases cash flows are **stochastic** 【许多现金流与市场指数挂钩，因此表现为一定的随机性，需要利率模型来预测。】
- ✓ ② 考虑信用风险调整：cash flows are adjusted to consider **credit risks** 【需要使用信用模型考虑违约风险及其相关性】
- ✓ ③考虑流动性期权调整：cash flows are adjusted to account for **liquidity options** 【使用行为模型处理特定银行产品（如活期存款、信用额度使用和抵押贷款提前还款）的现金流变化。】
- ✓ ④考虑新业务增加的现金流：cash flows originated, by **new business** increasing the assets should be included
- ✓ ⑤考虑到期负债的续期和新债券发行：the **rollover of maturing liabilities**, by similar or different contracts, and new bond issuances (which could also be included in the new business category) to fund the increase in assets, have to be taken into account.
- **Treasury Department 的任务：**
 - ✓ 监控 TSECF 和 TSECCF，确保 TSECCF 始终为正。
 - ✓ 理想状态是正现金流能够覆盖负现金流，但由于以下两个原因，这种情况不能总是实现：
 - 许多现金流在金额和时间维度上是随机 **stochastic** 的，TSECF 和 TSECCF 仅预测预期值。→→→ 解决办法 1：考虑现金流的区间范围，见稍后知识 “Cash flows at risk and the term structure of liquidity at risk”
 - 资产和负债到期的时间分布可能导致**短期内累积现金流为负**→→→ 解决办法：需要司库部门介入进行管理，**Liquidity generation capacity (LGC)**

2) Liquidity Generation Capacity(LGC)流动性生成能力

- Liquidity generation capacity (LGC) 流动性生成能力是银行应对 TSECCF 负值的主要工具。
- LGC: The ability of a bank to **generate positive cash flows**, beyond contractual ones, from the sources of liquidity available in the balance sheet

and off the balance sheet at a given date. 【本质上，LGC 就是通过银行的表内外资产，整出一些正的现金流】

- 两种方式
- ✓ ① **Balance sheet expansion** with secured or unsecured funding. 【Balance sheet expansion 资产负债表扩张，通过借债的方式获取流动性，本质就是借钱，其实对应的就是负债流动性管理】
 - ✧ borrowing through an increase of deposits, typically in the interbank market 通过银行间市场借钱
 - ✧ withdrawal of credit lines: 银行可以在其他机构那里借到钱
 - ✧ issuance of new bonds 发行新债券
- ✓ ② **Balance sheet shrinkage** is operated by **selling assets**, starting from more liquid ones such as Treasury bonds, corporate bonds and stocks
 - ✧ Balance sheet shrinkage 资产负债表缩水，本质上就是卖资产，其实对应的就是储存流动性管理
- ✓ 补充方法: balance sheet neutral 资产负债表中性: Repo transactions can also be considered separately from the other cases and labelled as "balance sheet neutral". 回购被认为是“资产负债表中性”的，因为回购不会在长期内改变资产负债表的总量和结构，只是在短期内产生流动性影响。
- **The term structure of LGC(TSLGC):** 不同时间点的 LGC
- The term structure of cumulated LGC(TSCLGC): Similar with TSECCF，是累计值
 - LGC 和 TSCLGC 是期望值，实际情况中，现金流的时间和数量通常是随机 **stochastic** 的
- Building the TSLGC can be very **difficult**:
 - ✓ the assumptions made for a given period affect other periods. 一个时期的假设会影响到其他时期
 - ✓ the TSLGC is **intertwined with the TSECF** (and thus the TSECCF)
 - 司库部门卖出 100 元的债券，获得流动性→影响 TSLGC
 - 但债券会有 coupon，属于 TSECF→影响 TSLGC
 - ✓ most of the problems in building the TSLGC are caused by unencumbered assets 未被质押的资产。【未被质押的资产在构建 TSLGC 时会增加复杂性，因为银行需要管理这些资产的流动性，并考虑它们受到的各种风险影响。】

● **The term structure of available assets (TSAA)可用资产的期限结构【构建 TSLGC 比较困难，进而可以使用 TSAA】**

- ✓ available assets: 可被抵押的资产
- ✓ The TSAA only shows how many single securities, or all of them, are available for inclusion in LGC.
- ✓ According to the type of operation (selling or repo) the price and/or the haircut are factors that need to be considered.

3) Case Study: 几类交易的 TSECF, TSECCF, TSAA 和 TSLGC 【要求自己知道表格数据怎么来的，同时一些定性结论需要掌握.这个是原版书例子，但是有些数据有问题，已经改过来了】

- ✓ ① Repo and reverse repo
- ✓ ② Buy/ sellback and sell/buyback
- ✓ ③ Lend and borrow
- 题干背景：
- ✓ The bank buys a bond at time 0 for a notional amount of 1,000,000 at a price of 98.50; the payment is settled after 3 days, when the bond's possession also passes to the bank. The bond pays a semiannual coupon of 10% p.a. and it expires in 2 years. 【Haircut=15%，指的是如果把这个债券抵押出去，100元只能借到 85 元】

Time	Operation	TSECF	TSECCF	TSAA	TSLGC	Price	Haircut (%)
0	Buy					99.85	15
0.01	Settlement	-998,500	-998,500	1,000,000		99.85	15
0.25				1,000,000		99.85	15
0.5	Coupon	50,000	-948,500	1,000,000		99.85	15
0.75				1,000,000		99.9	15
1	Coupon	50,000	-898,500	1,000,000		99.9	15
1.25				1,000,000		99.9	15
1.5	Coupon	50,000	-848,500	1,000,000		99.95	15
1.75				1,000,000		99.95	15
2	Coupon + Reimbursement	1,050,000	201,500	-		100	15

① **Repo and reverse repo**

- ✓ Repo 是融资，是递交抵押品，是司库部门发起，影响 TSAA 和 TSLGC;

- ✓ **reverse repo** 是投资，是收到抵押品，不是司库部门发起，通常由业务部门发起，所以不会影响 TSAA 和 TSCLGC

- ✓ Assume the bank repoes the bond after 3 months for a notional amount equal to 500,000 and a period of 6 months. Given the price and the haircut of the bond, and keeping accrued interest in mind, the amount of cash received by the bank is: 【在 0.25 年时，把 500000 面值的债券抵押出去，期限是 6 个月】

$$500,000 \times (99.85\% + 10\% \times 0.25) \times (1 - 15\%) = 434,988$$

- ✧ 在 0.25 时刻，银行把面值是 500,000 的债券，作为回购的抵押品，抵押出去，借到 434,988。

- ✓ The bank **pays 9%** as interest on this repo transaction, so that the terminal price paid when getting the bond back is:

$$434,988 \times (1 + 9\% \times 0.5) = 454,562$$

- ✧ 在 0.75 时刻，用 454,562 把债券买回来。
- ✧ 454,562 分为两部分：先把 TSCLGC(434,988) 归为 0 和多余/不足部分 -19,574 计入 TSECF。

- ✓ The difference $454,562 - 434,988 = 19,574$ enters the TSECF on the date at the end of the repo.

- ✓ 【说明：在 0.5 时刻、1 时刻、1.5 时刻，coupon 不变都是 50000。谁拥有债券，债券发行人会把 coupon 给到谁】

● Transaction: reverse repo

- ✓ After **1.25 years** the bank deals a 6-month reverse repo on this bond for a notional of 500,000. The price it pays to deliver the bond at inception is:

$$500,000 \times (99.9\% + 10\% \times 0.25) \times (1 - 15\%) = 435,200$$

- ✧ 在 1.25 时刻，收到抵押品，面值为 500,000 的债券，支付 435,200
- ✧ 支付：TSECF 为 -435,200；拿到抵押品：TSAA 变为 1,500,000

- ✓ At the end of the reverse repo contract, assuming the interest rate paid by the counterparty is **11%**, the inflow received by the bank is:

$$435,200 \times (1 + 11\% \times 0.5) = 459,136$$

- ✧ 在 1.75 时刻，给出抵押品，收到 459,136
- ✧ 收到：TSECF 为 459,136；给出抵押品：TSAA 变为 1,000,000

Time	Operation	TSECF	TSECCF	TSAA	TSCLGC	Price	Haircut (%)
0	Buy					99.85	15
0.01	Settlement	-998,500	-985,000	1,000,000		99.85	15
0.25	Repo		-985,000	500,000	434,988	99.85	15
0.5	Coupon	50,000	-985,000	500,000	434,988	99.85	15
0.75	End repo	-19,574	-968,074	1,000,000	-	99.9	15
1	Coupon	50,000	-918,074	1,000,000		99.9	15
1.25	Reverse repo	-435,200	-1,353,274	1,500,000		99.9	15
1.5	Coupon	50,000	-1,353,274	1,500,000		99.95	15
1.75	End reverse repo	459,136	-844,138	1,000,000		99.95	15
2	Coupon + Reimbursement	1,050,000	1205,862			100	15

② Buy/sellback and sell/buyback

● Transaction: buy and sellback 先买再卖【买债券，非司库部门发起】

- ✓ Assume that after 3 months the bank buys 400,000 bonds and sells it back after 6 months at the forward price. 【在 0.25 时刻买了面值是 400000 的债券，六个月后卖出】

- ✓ 0.25 时刻，买债券(不需要考虑 haircut)，支付：

$$400,000 \times (99.85\% + 10\% \times 0.25) = 409,400$$

◇ TSECF: -409,400

◇ TSAA: 1400000

- ✓ 0.75 时刻，债券价格是 99.9。卖出债券，收到：

$$400,000 \times (99.9\% + 10\% \times 0.25) = 409,600$$

◇ TSECF: 409,600

◇ TSAA: 1000000

● Transaction: sell and buyback 先卖再买

司库部门发起

- ✓ Assume that a sell/buyback of the bond starting after 1 year and 3 months (1.25 years) and terminating after 6 months (1.75 years) with 300,000 notional. 【1.25 时刻卖面值为 300000 的债券，1.75 时刻买债券】

- ✓ 1.25 时刻：The price received by the bank is:

$$300,000 \times (99.90\% + 10\% \times 0.25) = 307,200$$

◇ TSCLGC: 307,200

◇ TSAA: 700000

✓ 1.75 时刻: 买回这个债券, 支付:

$$300,000 \times (99.95\% + 10\% \times 0.25) = 307,350$$

$$307,350 - 307,200 = 150$$

■ 307,350 分为两部分: 先把 TSCLGC(307,200) 归为 0 和多余/不足部分 -150 计入 TSECF。

■ The TSAA increases back to 1,000,000.

【买入/卖出债券, 其 coupon 金额会根据面值变化而变化】

Time	Operation	TSECF	TSECCF	TSAA	TSCLGC	Price	Haircut (%)
0	Buy					99.85	15
0.01	Settlement	-998,500	-985,000	1,000,000		99.85	15
0.25	Buy	-409,400	-1,407,900	1,400,000		99.85	15
0.5	Coupon	70,000	-1,337,9400	1,400,000		99.85	15
0.75	Sellback	409,600	-928,300	1,000,000		99.9	15
1	Coupon	50,000	-878,300	1,000,000		99.9	15
1.25	Sell		-878,300	700,000	307,200	99.9	15
1.5	Coupon	35,000	-843,300	700,000	307,200	99.95	15
1.75	Buyback	-150	-843,150	1,000,000		99.95	15
2	Coupon + Reimbursement	1,050,000	206,850			100	15

③ Lend and borrow

● Lend/borrow 债券, 不影响 coupon

● Transaction: lend and borrow

✓ 借出: the bank lends 500,000 of the bond after 3 months for a period of 6 months. 【在 0.25 时刻, 借出面值为 500000 的债券, 期限是 6 个月】

◇ TSECF: 0, 借出去没有现金流的变化

◇ TSAA: 500,000 【借出去 500000, 还剩下 500000】.

✓ After 6 months the bond is returned to the bank (the TSAA increase) and the bank receives a fee for the lending, which we assume equal to 3% p.a. 【别人还债券, 并且还钱】

$$\text{还钱: } 500,000 \times (3\% \times 0.5) = 7,500$$

◇ TSAA: 1000000 【还债券】

◇ TSECF: 7,500

✓ 借入：在 1.25 时刻，借入期限是 6 个月的，面值为 300,000 的债券。

■ 在 1.75 时刻，银行还债券，还钱。

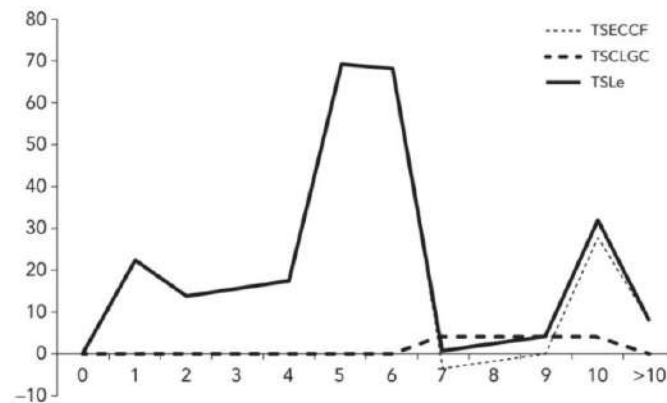
还钱： $300,000 \times (3\% \times 0.5) = 4,500$

还债券：TSAA=100000

Time	Operation	TSECF	TSECCF	TSAA	TSCLGC	Price	Haircut (%)
0	Buy					99.85	15
0.01	Settlement	-998,500	-998,500	1,000,000		99.85	15
0.25	Start lending		-998,500	500,000		99.85	15
0.5	Coupon	50,000	-948,500	500,000		99.85	15
0.75	End lending	7,500	-941,000	1,000,000		99.90	15
1	Coupon	50,000	-891,000	1,000,000		99.90	15
1.25	Start borrowing		-891,000	1,300,000		99.90	15
1.5	Coupon	50,000	-841,000	1,300,000		99.95	15
1.75	End borrowing	-4,500	-845,500	1,000,000		99.95	15
2	Coupon + Reimbursement	1,050,000	204,500	—		100.00	15

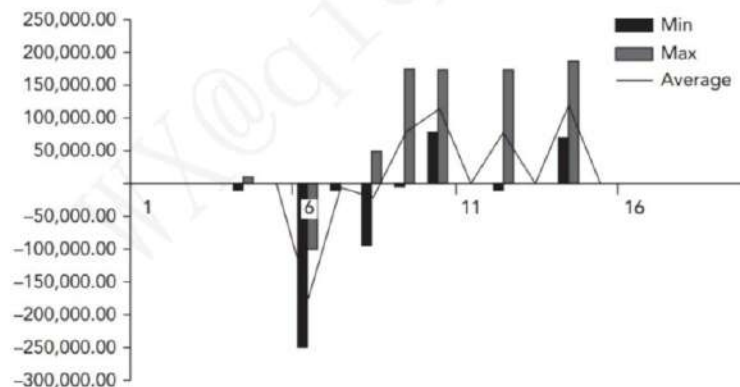
5. The term structure of expected liquidity

- The **term structure of expected liquidity (TSLe)** is basically a combination of the TSECCF and the TSCLGC. 【司库部门在流动性监控中扮演的角色】
- 逻辑关系：司库部门并不太关注 TSECF，是正是负，关系不大；比较关注 TSECCF
 - ✓ 如果 TSECCF 小于 0，司库部门介入，流动性生成能力→TSCLGC
- TSLe 涵盖所有可能由日常业务活动 ordinary business activity、新业务 new business、流动性政策 the liquidity policy operated 和应对负累积现金流的措施所产生的预期现金流。
 - ✓ **term structure of expected liquidity (TSLe)=Causes of liquidity+Sources of liquidity=TSECCF+TSCLGC**
- The TSL must always be positive if the financial institution has to be solvent all the time. 保持正的 TSLe 对于确保金融机构持续的偿付能力至关重要



6. Cash flows at risk and the term structure of liquidity at risk

- The fact that **cash flows are stochastic** suggests that some other measure related to its volatility. 【考虑现金流的随机性，引入其他指标，即区间估计，估计现金流数量可能发生的区间范围】
- 步骤：
 - ✓ 使用情景分析或模拟，找到现金流的分布
 - ✓ 找到期望值(CF_e),也就是现金流的预期值→TSECF
 - ✓ α 置信水平下的最大现金流和最小现金流: CF_{α}^{+} 和 $CF_{1-\alpha}^{-}$
- $TSCF_{\alpha}^{+}$: the upper bound of the term structure of stochastic cash flows
- $TSCF_{1-\alpha}^{-}$: the lower bound of the term structure of stochastic cash flows



● Cash flows at risk

- ✓ The positive and negative cash-flow-at-risk, defined respectively as:
 - ✧ positive cash-flow-at-risk: $CFaR_{\alpha}^{+} = CF_{\alpha}^{+} - CF_e$
 - ✧ negative cash-flow-at-risk: $CFaR_{\alpha}^{-} = CF_{\alpha}^{-} - CF_e$

Chapter 8 The Failure Mechanics of Dealer Banks(☆)

1. Large Dealer Banks 业务

- ✓ traditional commercial banks 传统商业银行：吸收存款发放贷款
- ✓ Dealer banks 交易商银行：在传统商业银行业务基础上，还有以下功能（这些功能也使得交易商银行面临比较大的风险）：
 - ✧ Securities Dealing, Underwriting, and Trading 证券交易、承销和交易：在一级市场中，交易商银行在证券发行人和投资者之间作为中介；在二级市场中，在投资者之间作为中介。交易商银行也参与自营交易，利用自身资本进行投机。
 - ✧ Over-the-Counter Derivatives 场外交易衍生品。
 - ✧ Prime Brokerage and Asset Management 主经纪业务和资产管理：为对冲基金和其他大型投资者提供一系列服务，包括管理证券持有、清算、现金管理、证券借贷、融资和报告。
 - ✧ Off-Balance-Sheet Financing 表外业务
- ✓ 交易商银行面对的主要风险：范围不经济（**diseconomies of scope**）。
 - ✧ 范围不经济是指在某些情况下，扩大公司的业务范围（例如同时管理多种类型的风险或业务）并不会带来预期的效率提升，反而可能导致效率下降。
 - ✧ 这里指的是在风险管理和公司治理方面，交易商业务范围过于广泛或复杂，可能无法有效地管理和控制风险。

2. Failure Mechanics of Dealer Bank

- 交易商银行在面临偿付能力质疑时，其有些客户会逃离，会导致银行的流动性快速枯竭，最终可能引发破产和系统性金融危机。具体可能逃离客户：
 - ✓ The Flight 逃离 of Short-Term Creditors → 银行挤兑 bank run
 - ✓ The Flight of Prime Brokerage Clients 主经纪客户的流失
 - ✓ When Derivatives Counterparties Duck for Cover 衍生品对手方的避险
 - ✓ Loss of Cash Settlement Privileges 失去现金结算特权

3. Policy Responses 政策回应

- ✓ higher capital requirements(GSIBs)更高的资本要求（全球系统重要性银行，GSIBs）
- ✓ lender-of-last-resort financing by central banks 中央银行作为最后贷款人提供融资
- ✓ the problems of short-term tri-party repos can be solved by central bank insurance of tri-party repo transactions and an "emergency bank 应急银

行," to be financed by repo market participants, that could manage the orderly unwinds of repo positions of weakened dealers. 在应对短期三方回购 (triparty repo) 带来的不稳定性时, 有两种政策措施可以缓解这些问题: 中央银行对三方回购交易提供保险和应急银行

- ✓ the flight of OTC derivatives counterparties can be lowered by central clearing. 场外交易 (OTC) 衍生品对手方逃离风险可以通过中央清算来降低
- ✓ Dealer banks could be given regulatory incentives or requirements to issue forms of debt(COCOs) that, contingent on stipulated distress triggers, convert to equity. 发行 COCOs 来进行风险管理

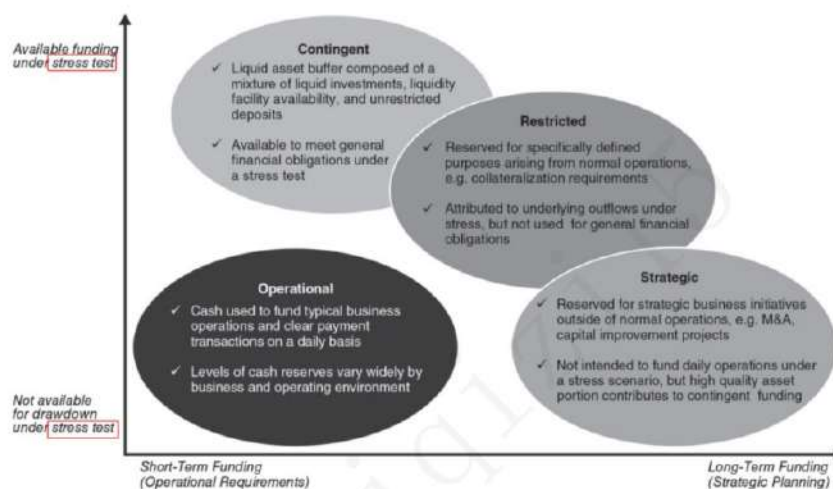
Chapter 9 Liquidity Stress Testing 流动性压力测试(☆)

在次贷危机之前，流动性很容易获得且成本低廉，许多银行虽然表面上资本充足，但无法确保在金融市场长期失衡的情况下保持充足的流动性。次贷危机突显了审慎管理流动性风险的重要性。

流动性压力测试目的：确定机构在压力情景下必须持有的流动性。

1. Contingent Liquidity

- 考虑两个维度：压力情况下资金的可得性，融资时间
- Utilize liquidity for four purposes: operational, restricted, contingent, and strategic.



- ✓ 运营流动性 Operational liquidity: 每天业务所需的现金，确保支付交易的有序清算，但不可用于流动性压力测试中的财务义务。
- ✓ 限制流动性 Restricted liquidity: 只能用于特定。
- ✓ 应急流动性 Contingent liquidity: 应急流动性就是在金融压力情境中，确保你有足够的高质量资产可以迅速变现。衡量应急流动性是压力测试的主要目标。
- ✓ 战略流动性 Strategic liquidity: 用于满足未来业务需求的现金，如收购或资本支出计划，但可转用于应急流动性需求。

2. Estimating Contingent Liquidity

- 估计压力情形下的应急流动性，需要包括以下几个模块：
 - ✓ 流动资产缓冲 (Liquid Asset Buffer): 机构当前所持有的应急流动性资产。模型需要清楚定义哪些证券可以被纳入流动资产缓冲，包括这些证券的市场和操作特性。
 - ✓ 压力情境下的资金流出 (Stressed Outflows): 预计在压力情境下会发生的资金流出。模型需要定义各种流出类型，例如零售存款流出、无担保融资流出等。

- ✓ 压力情境下的资金流入（Stressed Inflows）：预计在压力情境下的一些资金流入。模型需要考虑流入的来源，如贷款还款。
- ✓ 压力情境下的流动资产缓冲（Stressed Liquid Asset Buffer）：表示在扣除压力流出和流入后的流动资产缓冲的净值



3. Design of the model

● Organizational scope 模型涉及框架

- ✓ 流动性压力测试应从公司的整体层面开始，但也可以在母公司、子公司、业务线以及具体业务单元等层面进行单独测试。
- ✓ Liquidity transfer restrictions 流动性转移限制
 - foreign exchange controls may inhibit the conversion of foreign currency in off-shore legal entities 母公司与其海外子公司之间的流动性限制
- ✓ Currency 外汇问题
 - 外汇流动性可能受到影响，将外币转换为本币可能受到限制。
- ✓ Regulatory jurisdiction 监管环境
 - various regulatory oversight regimes

● Scenario development

- ✓ 基本情景（Baseline Scenario）最一般的情景。
- ✓ 情景开发（Scenario Development）
 - 历史情景（Historical Scenarios）：基于历史数据构建的情景。
 - 前瞻性（假设性）情景（Forward-Looking (Hypothetical) Scenarios）

● 假设（Assumptions）

- ✓ 投资组合折扣（Investment Portfolio Haircuts）：评估投资组合在压力情况下的折扣。
- ✓ 存款流出（Deposit Outflows）：
- ✓ 无担保批发融资（Unsecured Wholesale Funding）：通常在压力情况下大幅减少。
- ✓ 担保要求（Collateral Requirements）：压力情况下所需的额外担保。
- ✓ 其他应急负债（Other Contingent Liabilities）：例如担保。

- ✓ 业务缩减（Business Dial Back）：考虑在压力情况下减少流动性消耗的能力。
- 输出（Outputs）
 - ✓ 压力测试假设（Stress Testing Assumptions）。
 - ✓ 流动性位置指标（Liquidity Position Metrics）：流动性覆盖率（LCR）等。
 - ✓ 未来流动性头寸指标（Prospective Liquidity Position Metrics）
 - ✓ 资本和绩效指标（Capital and Performance Metrics）
- 治理与控制（Governance and Control）
 - ✓ Asset-liability committee 资产负债委员会（ALCO）：资产负债委员会通常负责流动性压力测试框架的
 - ✓ 三道防线（Three Lines of Defense）：
 - 司库部门（Treasury Unit）：作为第一道防线，通常负责流动性压力测试模型的过程。
 - 独立风险管理功能（Independent Risk Management Function）：作为第二道防线，负责提供流动性的独立监督。
 - 内部审计（Internal Audit）：作为第三道防线，定期审查流动性压力测试框架、程序和控制，确保符合政策、监管和控制要求。
 - ✓ 模型风险管理（Model Risk Management）：负责提供独立的验证。
- 与其他风险模型的集成（Integration of Liquidity Stress Testing with Related Risk Models）。理论上，机构应维持一个全面的风险模型，评估在共同情景下对流动性、资本和资产负债结构的影响
 - ✓ 流动性压力测试与资本压力测试（Liquidity Stress Testing and Capital Stress Testing）
 - ✓ 流动性压力测试与资产负债管理（Liquidity Stress Testing and Asset Liability Management）
 - ✓ 流动性压力测试与资金转移定价（Liquidity Stress Testing and Funds Transfer Pricing）【funds transfer pricing 资金转移定价见后面章节】

Chapter 10 Liquidity Risk Reporting and Stress Testing(☆)

1. Liquidity Risk Reporting

- Deposit Tracker Report (存款跟踪报告)
 - 存款跟踪报告提供当前存款规模以及未来存款水平的预测。报告按周和按月跟踪
- Daily Liquidity Report (每日流动性报告)
 - 每日流动性报告是一个详细的电子表格, 显示银行的流动和可交易资产及其负债的到期情况。它提供了银行日终流动性状况的总结。
- Funding Maturity Gap ("Mismatch") Report (融资到期缺口(错配)报告)
 - 融资缺口报告展示了按时间桶(time bucket)划分的到期缺口, 包括流动证券的调整, 并提供现金流生存期(survival horizon)报告。
- Funding Concentration Report (融资集中度报告)
 - 融资来源集中度报告显示银行在不同资金来源的集中程度, 包括大型存款集中报告。
- Undrawn Commitment Report (未使用承诺报告)
 - 未使用承诺报告展示了未使用的流动性和融资额度。
- Liability Profile (负债结构报告)
 - 负债结构报告展示了银行各种负债类型的占比。
- Wholesale Pricing and Volume (批发定价和数量)
 - 分析银行的融资成本和融资组成, 尤其关注融资收益曲线和产品结构。
- Summary and Qualitative Reports (总结和定性报告)
 - 高管层的流动性报告应为一页的关键流动性指标摘要, 每月提供, 特殊情况下可增加频率。
- Reporting Frequency (报告频率)
 - 主要流动性报告的频率由监管机构规定

2. Stress Test Reports

- 形式有两种
- ✓ cash flow survival report 现金流生存报告: 主要的压力测试输出结果。
 - 在压力情况下银行的现金流生存期, 以及在采取缓解措施(如出售证券或利用应急融资来源)后的变化。
- ✓ line-by-line stress test result report 逐行压力测试结果报告: 按季度或根据监管要求生成, 显示各个压力冲击对流动性比率的影响及其发生概率。报告包括
 - 流动资产减少、负债减少、外汇错配、综合冲击等类别。

■ 下图是流动资产减少、负债减少的情况

Stress Tests – Individual Shocks			Sight – 8 Day	Sight – 1 Month	Probability	Impact
Reduction in Liquid Assets						
Change in repo criteria	Light	Rating category 1 notch downgrade	8.46%	1.35%	50%	30
	Moderate	Rating category 2 notch downgrade	2.34%	0.12%	20%	70
	Severe	Rating category 3 notch downgrade	-15.2%	-18.2%	1%	90
Mark-to-market reduction in value of assets	Light		8.46%	1.35%	60%	20
	Moderate		2.34%	0.12%	40%	30
	Severe		-15.2%	-18.2%	5%	70
Increased haircut on assets	Light		8.46%	1.35%	70%	25
	Moderate		2.34%	0.12%	30%	45
	Severe		-15.2%	-18.2%	8%	80
Unavailability of repo facilities	Severe	Treat all marketable securities as illiquid (i.e., allocate to final legal maturity time buckets)	-15.2%	-18.2%	5%	100
Decrease in Liabilities						
Withdrawal of customer deposits	Light	Reduce customer deposits by 5%, replace with o/night funding	8.46%	1.35%	70%	20
	Moderate	Reduce customer deposits by 10%, replace with o/night funding	2.34%	0.12%	15%	30
	Severe	Reduce customer deposits by 15%, replace with o/night funding	-15.2%	-18.2%	5%	40
Withdrawal of corporate deposits	Light	Reduce Local Authority deposits by 25%, other deposits by 10%, replace with overnight funding	8.46%	1.35%	5%	20
	Moderate	Reduce Local Authority deposits by 50%, other Corporate Banking deposits by 35%, replace with overnight funding	2.34%	0.12%	2%	70
	Severe	Reduce Local Authority deposits by 100%, other Corporate Banking deposits by 70%, replace with overnight funding	-15.2%	-18.2%	1%	100
Withdrawal of intragroup deposits	Light	Reduce net group liability to EUR500 mm, replace with overnight funding	8.46%	1.35%	100%	5

Chapter 11 Contingency Funding Planning 应急融资计划(☆☆)

1. 基本概念

- **Contingency Funding Planning** 危机时，如何融得资金的一套方案
- contingency funding plan (CFP)：流动性压力测试，关注的是在危机时需要获得多少资金；而 CFP 指的是危机时，如何融得资金的一套方案
- 应急流动性事件的分类：
 - ✓ 低影响、高概率事件 low-impact, high-probability events: 作为日常业务中的流动性管理活动来处理。【正常需要管理的资金，和这里的 CFP 没有关系】
 - ✓ 高影响、低概率事件 high-impact low-probability events: 需要通过 CFP 来应对。

2. Key Design Considerations

- ✓ ① Aligned to business and risk profiles 与业务和风险特征对齐
 - CFP 应根据机构的具体业务和风险特征进行量身定制。另外，CFP 必须与机构的风险偏好一致，并随着业务环境和内部策略的变化进行定期更新。
- ✓ ② Integrated with broader risk management frameworks 与风险管理框架整合
 - CFP 应与机构的整体风险管理框架（如 ERM、资本管理和业务连续性计划）整合。这种整合确保 CFP 的连贯性和有效性。
- ✓ ③ Operational and actionable, but flexible playbook 剧本/方案
 - include a menu of possible contingency actions that management can undertake in different stress scenarios and at graduated levels of severity. CFP 应提供针对各种压力情景和严重程度的应急行动清单。
- ✓ ④ Inclusive of appropriate stakeholder groups 包含适当的利益相关者
 - including various management committees (e.g. ALCO, risk and capital committee, investment committee), business units, finance, treasury, risk, operations and technology 有效的 CFP 涉及各种利益相关者，包括管理委员会、业务部门和操作团队。确保了 CFP 的全面性，并在危机期间有效地协调。

- ✓ ⑤ Supported by a communication plan 配备沟通计划

3. Framework and Building Blocks 【Building Blocks 基本模块】

- ✓ ① Governance and oversight 治理和监督

- 明确角色和责任，制定强有力的沟通策略，并通过定期测试和演练来强化计划。
 - **Liquidity crisis team** 流动性危机团队（LCT）：LCT 负责持续监控机构的流动性状况。一般情况下，LCT 负责设计应急资金计划（CFP）并提交给高级管理团队进行审查和批准。
- ✓ ② Scenarios and liquidity gap analysis 情景分析与流动性缺口分析
- The liquidity stress testing scenarios will cover both **systemic (general market) and idiosyncratic (institution-specific) risks** and address both **market (asset) liquidity and funding liquidity**, over **short-term and prolonged** stress periods. 【流动性压力测试情景，包括系统性风险和非系统性风险，市场流动性和融资流动性，涵盖短期和长期压力情况。
 - CFP 应提供应对危机的战术机制，并提供应对措施。
- ✓ ③ Contingent actions 应急措施
- A list of actions such as selling business or business units, curtailing discretionary spending and expenses, etc..
- ✓ ④ Monitoring and escalation 监控与升级
- ✓ ⑤ Data and reporting: CFP requires updating on a regular basis.

Chapter 12 Managing and Pricing Deposit Services(☆☆☆)

存款 Deposits 是大多数银行的主要资金来源。本节内容涉及的存款种类，主要是美国市场，和国内市场会有不同。

1. Types of Deposits

- Transaction (Payments or Demand) Deposits 交易（支付或活期）存款
 - ✓ Noninterest-Bearing Transaction (Demand) Deposits 无息交易（活期）存款
 - ✧ 这些存款不赚取利息，但提供支付服务、资金保管和交易记录。
 - ✧ 用户：企业客户
 - ✓ Interest-Bearing Transaction Deposits 有息交易存款，有如下种类
 - ✧ Negotiable Order of Withdrawal (NOW) Accounts 可转让提款命令账户（NOW 账户）
 - 可用于交易的有息储蓄存款，类似于支票账户。
 - 用户：个人和非营利机构。
 - ✧ Money Market Deposit Accounts (MMDAs) 货币市场存款账户（MMDAs）
 - 短期存款，提供有竞争力的利率，允许用户开具支票，但有一定的限制。这些限制可能包括每月允许开具支票的次数、每次开具支票的金额限制等。
 - 用户：企业和个人。
 - ✧ Super NOWs (SNOWs) 超级 NOW 账户（SNOWs）
 - 类似于 NOW 账户，但允许用户开具支票，并且不对每月开具支票的次数进行限制。
 - 用户：个人和非营利机构。
 - ✓ Mobile Check Deposit 移动支票存款：允许客户使用智能手机拍照存入支票。
- Non-Transaction (Savings or Thrift) Deposits 非交易（储蓄或节俭）存款
 - ✓ Passbook Savings Deposits 存折储蓄存款
 - 小额储蓄存款，提款权限无限，通常记录在实体存折中。
 - 用户：个人、非营利组织、政府和企业。
 - ✓ Statement Savings Deposits 对账单储蓄存款
 - 通过计算机记录的储蓄账户，定期提供对账单。
 - 用户：普通客户。
 - ✓ Time Deposits (CDs) 定期存款（CDs）

- 具有固定到期日的存款，利率固定或波动。
- Types:
 - ◆ 可转让 CDs(Negotiable CDs): 在到期前可交易，通常超过 10 万美元。
 - ◆ 不可转让 CDs(Nonnegotiable CDs): 在到期前不可交易，通常为小额存款。
 - ◆ 创新型 CDs(Innovative CDs): 具有升息、阶梯利率、流动提款和指数联动收益等特点的 CDs 变种。
- Retirement Savings Deposits (IRAs, Keogh Plans) 退休储蓄存款 (IRAs, Keogh 计划)
 - ✓ 用于退休储蓄的税收优惠账户，包括 IRAs 和 Keogh 计划。
 - ✓ 用户：工资收入者、薪金个人。

2. Pricing Different Types of Deposits

- 一共 4 种方式，最后一种方式不在考纲范围内。
- ① **Cost plus profit** 成本加利润定价法 → 交易存款
 - ✓ 将存款服务的生产成本与计划利润结合起来，计算出单位存款服务的价格。
 - unit price charged** to the customer for each deposit service
 - = **operating expense** per unit of deposit service
 - + **estimated overhead expense** allocated to the deposit service function
 - + **planned profit margin** for each service unit sold
 - ✓ 单位价格 = 运营成本 + 分摊的间接费用 + 计划利润
 - ✓ 鼓励存款机构紧密地匹配价格和成本，消除了许多的免费服务。
 - ✓ 例子：Bank of the Bluegrass offers a demand deposit account that has a \$500 minimum deposit. The cost to service the account, including labor and computer time, is \$2.50 per month. The allocated overhead expense is \$1.25 per month, and the bank would like to build in a \$0.25 profit margin per account. Calculate the monthly fee the bank should charge the customer using the cost-plus pricing method.
 - Answer:
 - Monthly fee per account = \$2.50 + \$1.25 + \$0.25 = \$4.00 per month

② **Marginal cost** 边际成本定价法 → 非交易存款

- ✓ **Marginal cost:** the added cost of bringing in new fund 使用增加新资金的边际成本来设定存款利率，而不是使用历史平均成本。

Marginal cost 边际成本 = Change in total cost = New interest rate × Total funds raised at new rate - Old interest rate × Total funds raised at old rate = (新利率 * 新资金总量) - (旧利率 * 旧资金总量)

Marginal cost rate = Change in total cost / Additional funds raise

- ✧ Marginal cost 小于 Marginal revenue, 继续融资, 其放贷
- ✧ **Marginal cost = Marginal revenue, 利润最大化**
- ✧ Marginal cost 大于 Marginal revenue, 不融资, 不放贷

③ **Conditional pricing** 条件定价法

- ✓ 根据存款余额、交易次数或存款期限等因素，设定不同的费用。
 - 举例：假设银行 A 和银行 B 的定价策略如下：
 - ◆ 银行 A 对低于 \$600 的账户余额收取 \$5-\$10 的月费；不限次数。
 - ◆ 银行 B 对低于 \$500 的账户余额收取 \$3.5 的月费；超过 10 笔取款的账户每笔收费。
 - 客户根据账户余额和交易次数选择最适合的存款计划：
 - ◆ 若客户余额高于 \$600 且交易笔数较多，选择银行 A。
 - ◆ 若客户余额低于 \$500 且交易笔数较少，选择银行 B。
- ✓ checking accounts 支票账户的运用
 - 固定费用定价 (Flat-rate Pricing): 对账户收取固定费用或每次交易收取固定费用。
 - 免费定价 (Free Pricing): 账户和交易均不收取费用。
 - 有条件免费定价 (Conditionally Free Pricing): 账户费用根据存款余额的大小而变化，小额账户可能会收取费用，而大额账户则免费。

④ based on total customer relationship and choosing a depository 【根据与客户关系进行定价，不在考纲中】

3. **Challenges of Deposit Offering**

- Deposit insurance(FDIC) 联邦存款保险公司 (FDIC) 提供的存款保险。
- Disclosure 披露：法律要求银行提供详细的账户条款信息，包括最低余额、利息计算方法和费用结构。这增加了银行的行政负担。
- Overdraft Protection 透支保护：透支保护可能导致客户更频繁透支，增加财务风险

- Basic (lifeline) banking 基本（生命线）银行服务：银行是否履行一些社会责任

小知识：联邦存款保险公司（FDIC）提供的存款保险 Deposit Insurance

- 存款保险为存款人在银行破产或倒闭时提供保障。在美国，目前每位存款人在每家银行的保险覆盖上限为 250,000 美元。如果银行破产，保险公司会赔偿存款人在限额内的所有损失。
- 存款保险通常涵盖以下类型的账户：
 - ✓ Demand deposits.
 - ✓ Savings deposits.
 - ✓ NOW accounts.
 - ✓ Christmas club accounts.
 - ✓ Time deposits (CDs).
 - ✓ Money orders.
 - ✓ IRA and Keogh deposits 等
- 存款保险不包括以下项目：
 - ✓ 政府债券和证券
 - ✓ 共同基金份额
 - ✓ 保管箱内物品
 - ✓ 从保险机构被盗的资金

银行根据其风险水平和资本充足程度支付保险费。高风险银行需支付更高的保险费，以资助保险基金。

Chapter 13 Managing Non deposit Liabilities(☆☆)

- 大多数存款机构的传统资金来源是储户的存款。当存款不能带来足够的资金时，其中一个重要措施，就是主动负债，也就是这个 chapter 所涉及的非存款类负债。
- 近些年，非存款负债超越传统存款的增长，部分原因在于非存款负债具有更大灵活性和较少的监管，另外近年来还有些存款流失到其他的金融机构（如共同基金、保险公司、对冲基金和养老基金）手中。

1. Sources of Nondeposits

① Federal Funds Market ("Fed Funds") 联邦基金市场

- 知识点补充：
 - ✓ 联邦基金市场 (Federal Funds Market) 是银行之间进行短期借贷的市场。这个市场主要用于银行之间调节储备金，以满足法律要求和日常运营的需要。
 - ✓ 联邦基金指的是银行在联邦储备银行 (Federal Reserve) 或其他银行的存款。因为银行需要在每个营业日结束前保持一定量的准备金 (一般是法定准备金要求 legal reserve requirements)。如果一家银行在当天准备金不足或者临时资金需求，可以通过联邦基金市场借入资金。
- The most popular domestic source of borrowed reserves.
- three types of loan agreements
 - ✓ overnight loans 隔夜贷款
 - ✓ term loans 定期贷款
 - ✓ continuing contracts 持续合同
- ② Repurchase Agreements as a Source of Funds 【后面 chapter 详细介绍】
- ③ Borrowing from Federal Reserve Banks 从联邦储备银行借款→从央行借款

- ✓ Primary credit 初级信贷：短期贷款（通常隔夜），利率略高于联邦储备的目标利率。
- ✓ Secondary credit 次级信贷：利率较高，用于不符合初级信贷条件的机构，需接受联邦储备银行的监控
- ✓ Seasonal credit 季节性信贷：为小型和中型机构提供长期贷款，与市场利率挂钩
- Secondary credit generally posting the highest interest rate and seasonal credit the lowest.

- 当银行需要资金时，可以通过联邦储备银行的贴现窗口（Discount Window）借款。

④ **Advances from Federal Home Loan Banks(FHLB)** 从联邦家庭贷款银行借款

- 联邦住房贷款银行（Federal Home Loan Banks, FHLB）是由美国政府资助的金融机构，成立于 1932 年。FHLB 系统由 12 家地区性银行组成，为其成员（如银行、储蓄和贷款协会、信用社、保险公司）提供贷款和其他金融服务。FHLB 由联邦住房金融局（FHFA）监管，确保其运营的安全性和稳健性。
- Fully collateralized by home mortgages the maturities of FHLB loans range from overnight to more than 20 years, bearing either fixed or variable interest rates.

⑤ **Development and Sale of Large Negotiable CDs**

- This funding source is really a **hybrid account**: legally, it is a deposit, but, the negotiable CD is just another form of IOU(I owe you) in practice. 【是存款，但是具有流动性，可以在二级市场交易】
- There are four main types of negotiable CDs today:
 - ✓ **Domestic CDs** 国内 CD are issued by U.S. institutions inside the territory of the United States.
 - ✓ **Dollar-denominated CDs** 欧洲 CD: 是指在欧洲市场上发行的美元定期存单。通常由欧洲的银行发行，以美元计价。
 - ✓ **Yankee CDs**: 由美国境外的银行在美国市场上发行的定期存单，以美元计价。
 - ✓ nonbank savings institutions sell **thrift CDs**
- In the United States at least, legal reserve requirements are currently zero for CDs.

⑥ **The Eurocurrency Deposit Market** 欧洲货币存款市场

- Eurodollars are dollar-denominated deposits placed in bank offices outside the United States.
- Major banks and their large corporate customers practice **arbitrage** between the Euro and American CD markets.

⑦ **Commercial Paper Market**

- Commercial paper 商业票据是由知名公司发行的短期票据
- The notes are generally sold at a discount from face value through security dealers or through direct contact with the issuing company.

⑧ Long-Term Nondeposit Funds Sources 长期非存款资金来源 【特定项目】

- **Long-Term Nondeposit Funds:** 涉及一年以上的长期借款，如为特定项目而发行的抵押贷款、资本票据和债券（通常期限为 5 至 12 年），
- 长期资金来源是发行机构感知风险（特别是违约风险）的敏感晴雨表 barometer。

2. Chooses of Nondeposits

- 有很多非存款资金来源，管理层需要考虑两个问题：融多少？选择哪一(几)种？

- **Available funds gap** 可用资金缺口→衡量融多少问题

- ✓ The difference between current and projected outflows and inflows of funds yields an estimate of each institution's **available funds gap**

Available funds gap (AFG) = Current and projected loans and investments the lending institution desires to make - Current and expected deposit inflows and other available funds = 需要的资金量 - 已有的资金量

- ✓ 例如：suppose a commercial bank has new loan requests that meet its quality standards of \$150 million; it wishes to purchase \$75 million in new Treasury securities being issued this week and expects drawings on credit lines from its best corporate customers of \$135 million. Deposits and other customer funds received today total \$185 million, and those expected in the coming week will bring in another \$100 million. This bank's estimated available funds gap (AFG) for the coming week will be as follows (in millions of dollars):

- ✓ $AFG = (\$150 + \$75 + \$135) - (\$185 + \$100) = \75

- Five factors to consider→选择哪一(几)种？

- ✓ ① **relative costs**

- 有两种方法：historical average cost approach and the pooled-funds approach.

- ✓ ② **risk (volatility and dependability):** 需要考虑利率波动和信用可用性风险 credit availability risk

- 信用可用性风险 credit availability risk 则是借贷市场的紧张程度可能限制借款者的资金获得。

- ✓ ③ **length of time (maturity or term) :** 短期需求适合从联邦基金市场借款，长期需求则考虑长期债务。

- ✓ ④size of the institution: 大机构可以利用大额资金来源, 小机构则可能需要更灵活地考虑
- ✓ ⑤ Regulations limiting: 需遵守相关法规, 考虑对借款成本和风险的影响

3. Costs of Nondeposits

- ① The Historical Average Cost Approach 侧重使用历史数据
 - ✓ This approach for determining how much funds cost looks at the past. It asks what funds the financial firm has raised to date and what they cost.
 - ✓ 原版书例题:

Sources of Funds Drawn Upon	Average Amount of Funds Raised (millions)	Average Rate of Interest Incurred	Total Interest Paid for Each Funds Source (millions)
Noninterest-bearing demand deposits	\$100	0%	\$0
Interest-bearing transaction deposits	200	7%	14
Savings accounts	100	5%	5
Time deposits	500	8%	40
Money market borrowings	100	6%	6
Total funds raised = \$1,000			All interest costs = \$65

- A. Calculate the average interest cost of deposits and money market borrowings.
- ✧ Weighted average interest expense = All interest paid / Total funds raised
 $= \$65 / \$1000 = 6.5\%$
- B. Consider other operating costs, such as salaries and overhead, must also be covered. If these are an estimated \$10 million, calculate the breakeven cost rate on borrowed funds invested in earning assets (\$750 million): 【earning assets 生息资产, 也就是只有这 750million 能挣钱】
- ✧ Breakeven cost rate = (Interest + Other operating costs) / All earning assets
 $= (\$65 + \$10) / \$750 = 10\%$
- 【Breakeven cost rate 指的是让银行不赚不亏的利息成本。也就是如果发放贷款, 贷款利率至少应该 10%, 否则银行会有亏损】
- C. If the borrowing institution's stockholders and their after tax (35% tax rate) required rate of return is 12% and equity shareholders invest \$100 million in the institution, what is the weighted average overall cost of capital of the institution?
- ✧ Weighted average overall cost of capital = $\{65 + 10 + [12\% / (1 - 0.35)] \times 100\} / 750 = 12.5\%$

- ✧ **【Weighted average overall cost of capital:** 平均融资成本，既反映了项目本身的成本（利息成本、运营成本）又考虑了债权人和股东的成本，是一个综合的成本。也就是公司金融/理财中 WACC，FRM 没怎么讲过 WACC】
- ✧ Thus, 12.5 percent is the lowest rate of return over all fund-raising costs the borrowing institution can afford to earn on its assets.

② The Pooled-funds Approach 侧重预期，即未来数据

- ✓ This method of costing borrowed funds **looks at the future**: What minimum rate of return must be earned on any future loans and investments just to cover the cost of all new funds raised?

只有部分资金可以用来赚钱

Profitable New Deposits and Nondeposit Borrowings	Dollars of New Deposit and Nondeposit Borrowings (millions)	Portion of New Borrowings That Will Be Placed in New Earning Assets	Dollar Amount That Can Be Placed in Earning Assets (millions)	Interest Expense and Other Operating Expenses of Borrowing Relative to Amounts Raised	All Operating Expenses Incurred (millions)
Interest-bearing transaction deposits	\$100	50%	\$50	8%	\$8
Time deposits	100	60%	60	9%	9
New stockholders' investment in the institution	100	90%	90	13%	13
Total	\$300		\$200		\$30

A. Calculate the overall cost of new deposit **【新融资的成本】**

- ✧ Pooled deposit and nondeposit funds expense
=all expected operating expenses/all new funds expected
=\$30million/\$300million =10 percent

B. Suppose only two-thirds of these expected new funds (\$200 million out of \$300 million raised) will actually be available to acquire earning assets, calculate the hurdle rate of the deposite.

- ✧ Hurdle rate of return over all earning assets
=all expected operating costs/dollars available to place in earning assets
=\$30million/\$200million =15%

【Hurdle rate 就是能够接受的最低值】

Chapter 14 Repurchase Agreements and Financing(☆☆☆)

1. 回购基本概念

- A **repurchase agreement or repo** is a contract in which a security is traded at some **initial price** with the understanding that the trade will be **reversed at some future date at some fixed price**.

- 例如:

- ✓ 0 时刻: 公司 A 把抵押品(债券)给到 B, 融到 111772000 欧元; repo rate=0.23%, 期限是 92 天.
 - ✧ 111772000 也叫 initial price 或 Purchase Price₀
 - ✧ 债券实际价格 Security Price₀ 这里是未知, 正常而言, 债券的实际价格大于 111772000
- ✓ 1 时刻: 公司 A 用 repurchase price 把原来抵押品买回来。

$$\text{Repurchase Price} = \text{Purchase Price}_0 \times \left(1 + \frac{T}{360} \times \text{Repo rate}\right)$$

$$\text{Repurchase Price} = 111772000 \times \left(1 + \frac{92}{360} \times 0.23\%\right) = 111837697.10$$



Figure 14.1 The initiation of a repo trade.



Figure 14.2 The unwinding of a repo trade.

- ✓ A 方: seller of the security/ borrower (with the security serving as the collateral). 目的是融资
- ✓ B 方: buyer of the security/lender (with the higher sum representing principal plus interest). 目的是投资
- ✓ 资产债券看作 collateral。所以, 回购也可以看作是一个抵押贷款 collateralized loan

- 回购中，作为抵押品的资产可以有很多种类。不同种类、不同风险的抵押品其 haircut 也不同。

$$\text{Haircut} = \frac{\text{Security Price}_0 - \text{Purchase Price}_0}{\text{Security Price}_0}$$

- 回购根据回购期限分类：
 - ✓ 隔夜回购 Overnight repo: the term is one day → 最普遍
 - 通常把回购看成短期有担保的融资方式
 - ✓ 期限回购 Term repo: the term is more than one day.
 - ✓ Repo to maturity: An agreement lasting until the final maturity date
- Reverse repurchase agreement/Reverse repo 逆回购：由资金贷出方（借入证券）发起
- Repurchase agreement 回购：由资金借入方（借出证券）发起

2. 回购的原因

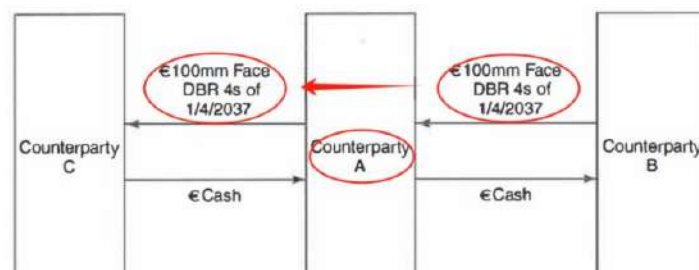
- ① to lend funds short-term on a secured basis → 目的，投资
- ② finance along position in a security → 目的，融资
- ③ to borrow a security in order to sell it short. → 目的，获得某个资产，然后卖出(对应逆回购)

① Repos and Cash Management (Repo Investors)

- ✓ 在不太牺牲流动性的同时获得短期利率，并且违约风险较低
- ✓ 主要回购投资者：
 - 货币市场基金
 - 以税收收入为来源的市政机构
 - 共同基金、保险公司、养老基金和一些非金融公司。
- ✓ 抵押品通常只接受最高信用质量(highest credit quality)的证券。

② Repos and Long Financing (Repo Borrowers)

- ✓ 金融机构是回购市场中典型的借款人
- ✓ Back-to-back repo trades 背对背回购交易：指的是连续回购交易。在“背靠背”回购交易中，交易方在一个交易完成后，立即使用所得的证券进行另一笔回购交易。



③ Reverse Repos and Short Positions

- ✓ 发起逆回购是为了借入债券。
- ✓ Repo transactions that require the delivery of a particular bond are called **special trades** and they take place at special collateral rates.

3. Repos During the Credit Crisis

● 贝尔斯登 (Bear Stearns)

- ✓ 贝尔斯登是一家投资银行，依赖于**短期融资**来支持其业务运营。
- ✓ 在 2008 年 3 月，贝尔斯登经历了一场“银行挤兑 bank run”。市场谣言 (market rumors) 导致客户、贷款人和交易对手对贝尔斯登失去信心。
- ✓ 特别是，**回购市场的贷款人拒绝续签 (roll over)**，即使是由**高质量抵押品支持的回购贷款**，这加剧了贝尔斯登的流动性危机，2008 年 3 月 16 日，摩根大通宣布以每股 2 美元的价格收购贝尔斯登。

● 雷曼兄弟 (Lehman Brothers)

- ✓ 雷曼兄弟 (Lehman Brothers) 成立于 1850 年，是美国历史最悠久的投资银行之一。在次贷危机前，雷曼兄弟大量投资于次级抵押贷款和相关证券。这些高风险资产在房地产市场下滑和次贷危机爆发后，迅速贬值，严重影响了雷曼兄弟的财务状况。
- ✓ 雷曼兄弟依赖于**三方回购 (tri-party repo)**市场来获得**短期融资**。在 2008 年初开始对手方 (例如：JPMorgan Chase) 要求雷曼兄弟提供更高的抵押品折扣 (即保证金)。
- ✓ 雷曼兄弟提供的许多**抵押品**被发现是**流动性差且难以估值**的结构性债务工具。这些抵押品的价值被高估，加剧了市场对雷曼兄弟资产质量的担忧。
- ✓ 2008 年 9 月 15 日，雷曼兄弟正式申请破产保护，成为美国历史上最大的一次破产案。

4. Collateral in Repo Transactions

- Repo trades can be divided into those using **general collateral (GC)** and those using **special collateral (SC)** or specials.
- ✓ 一般抵押品 (General Collateral, GC)
 - ✧ 一般抵押品 (General Collateral, GC): the lender of cash is willing to take any particular security. 借出方关心抵押品的整体质量，而不关心具体的证券或资产
 - ✧ 对应的 rate 为 **GC rate**，**回购中的抵押品越好，GC rate 越低**。反之亦然

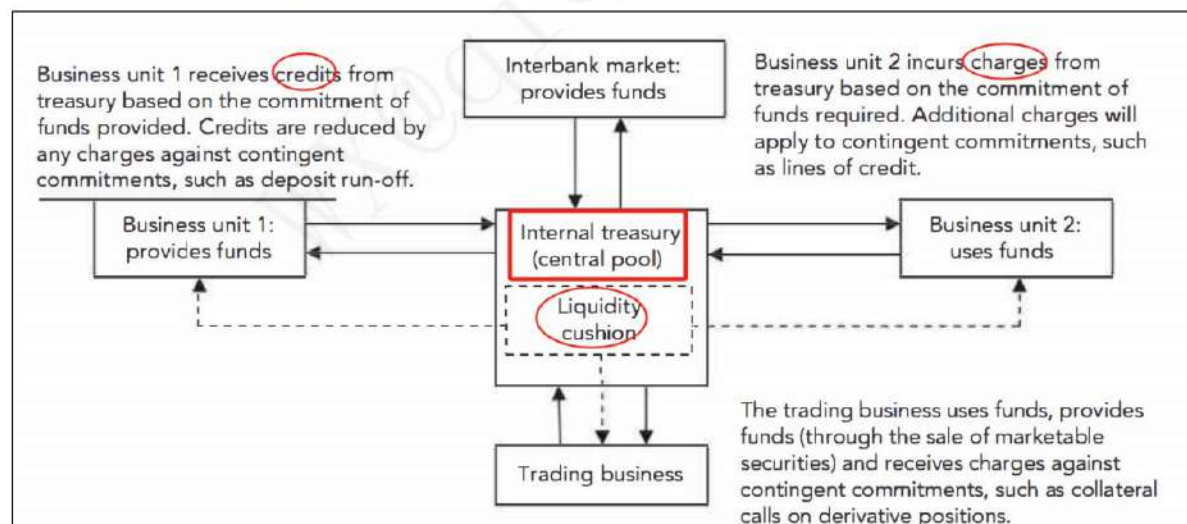
- ✧ GC rate 通常略低于联邦基金利率 FFR。
 - **fed funds-GC spread**: 联邦基金利率 FFR 与 GC 利率之间的差额。
 - Spread 变大: FFR 上升时, 或者 GC rate 下降。比如当美国国债变得稀缺时, 这个 Spread 会扩大 (GC 利率下降)
- ✓ 特殊抵押品 (Special Collateral, SC)
 - ✧ 特殊抵押品 (Special Collateral, SC): the lender of cash initiates the repo in order to take possession of a **particular security**. 【借出方比较关注抵押品本身】
 - ✧ 对应的 rate 为 SC rate。
- ✓ **Special spreads**: Differences between the GC rate and the SC rates for particular securities and terms
 - ✧ **FFR > GC rate > SC rate**
- ✓ Special spreads 的水平 and 波动是市场情绪的重要指标, 会随时间波动。但在拍卖周期 Auction Cycle 中有特定的模式。
 - 拍卖周期 (Auction Cycle) 是指美国政府按固定时间表进行国债拍卖的周期性过程。具体来说, 美国财政部会定期发行不同期限的国债, 如 10 年期国债每三个月进行一次拍卖。
 - ❖ On-the-run (OTR)(或者 Current issues): 最近发行的某一期限的债券。通常是最具流动性的。
 - ❖ Off-the-run (OFR): 除 OTR 债券外的所有其他发行的债券
 - 拍卖后 Special spreads 通常会变窄。这是因为新发行的 OTR 证券市场供应增加, SC rate 会上升 (在回购中, 这种 OTR 供给增多, 不是那么稀缺了, 投资者就会要求更高的回报)。
 - 在拍卖前 Special spreads 通常会变宽。
- 5. **Financing Value of A Bond Trade Special**
 - ✓ 在回购交易中交易 SC 时的融资优势【因为在 SC 中, 是资金借出者喜欢的资产, 因为对应的回购利率比较低, 存在融资优势比 GC】
 - ✓ Example:
 - ✧ The bond has a price of \$101.90, assuming no haircuts; Special spread is 0.22% and the repo term is 125 days; The DV01 of the bond is 0.085;
 - ✧ **Financing Advantage = $101.9 \times (125 \times 0.22\%) / 360 = \0.077**
 - ✧ $0.077 / 0.085 = 0.9$ basis points
 - ✧ Special Trade 带来的融资好处, 相当于市场利率下降 0.9BP

Chapter 15 Liquidity Transfer Pricing 流动性转移定价(☆☆☆)

背景引入：银行通过期限转换(Maturity transformation, 借短投长)来盈利，但这种操作使其容易受到流动性风险的影响。在全球金融危机之前，许多银行将流动性视为一种免费的资源，并未正确管理 LTP。不好的 LTP 导致了全球金融危机期间的流动性问题。

1. LTP Introduction

- LTP is a process that attributes the **costs, benefits and risks** of liquidity to respective **business units within a bank**. LTP 是银行管理中用于分配流动性成本、收益和风险的一种机制。
 - ✓ 说白了，LTP 帮助银行确定各个部门在使用和提供流动性时应该支付或获得的价格。
- **provides funds(提供流动性)的部门**：如存款业务部、资本市场部等→获得激励,例如下图中的 Business unit 1
- **uses funds (使用流动性)的部门**：如贷款部、投资银行部等→支付价格,例如下图中的 Business unit 2
- **流动性缓冲 (Liquidity Cushion)** 是指银行持有的高流动性的资产储备，应对紧急的流动性需求。即从提供流动性的部分支付更少的激励，从使用流动性的部门收取更高的价格，节流一部分的资金。



小知识：“credit”的几种含义

- 账户信用：在银行账户或其他金融账户中，“credit”表示账户余额的增加。
- 信贷额度：在贷款或信用卡账户中，“credit”指的是您可以借用的金额。
- 奖励或补偿：在**流动性转移定价 (LTP)**方法中，“credit”指的是对某项资金提供的**奖励或补偿**。具体来说，这通常指的是银行为吸引资金或流动性提供给客户的额外利益。
- 会计借方：在会计中，“credit”指的是账户的一种记录方式，用于记录负债、收入或资本的增加。

2. Governing LTP

- ✓ 制定 LTP 政策：Banks should have an **LTP policy** that defines LTP, states the purpose of LTP and provides some principles and/or rules to ensure LTP achieves its intended purpose. 明确 LTP 定义和目的，制定适用于所有重大流动性资金使用或提供的业务单位的政策
 - ✓ 集中管理：LTP should be **managed centrally**, within group treasury or a subsidiary treasury, and applied consistently across the group. LTP 应由集团司库部门集中管理
 - ✓ Trading book funding policies 交易账簿融资政策：银行需要对其交易活动进行细致的资金管理，确保每个交易台的风险和资金需求都能得到适当的管理和控制
 - ✓ 独立监督：Oversight of the LTP process should be provided by **independent risk and financial control personnel**. Senior management should also be involved in the LTP process. LTP 过程应由独立的风险和财务人员监督，高级管理层应参与定期会议
 - ✓ 先进的 LMIS 系统：To assist with performance assessments and decision-making, LMIS should be advanced enough to attribute, at a sufficiently granular level, all of the relevant costs, benefits and risks of liquidity to the appropriate business activities. LMIS 系统需足够先进，详细分配流动性成本、收益和风险到具体业务活动；流动性管理信息系统（Liquidity Management Information System, LMIS）
 - ✓ 薪酬调整：Remuneration practices should be more sensitive to the risks taken to generate profits. 银行在设定薪酬和奖金时，应更多地考虑风险因素，而不仅仅是利润
3. 实施 LTP 面临的挑战
- ✓ maintaining proper oversight to manage and monitor the process 管理和监控 LTP 过程，以确保流动性溢价定价正确。
 - ✓ reducing internal arbitrage opportunities 确保集中化运作，降低由于去中心化融资带来的套利机会。
 - ✓ ensuring the information system infrastructure is sufficient to produce high-quality useful reports in a timely manner 确保流动性管理信息系统（LMIS）能够及时生成高质量的有用报告。
 - ✓ ensuring that the remuneration policy for executives of business units and personnel properly account for liquidity costs and risks 确保高管薪酬政策的合理性

4. 风险定价

1) "Zero" Cost of Funds Approach →把流动性作为“免费”的，加剧期限错配(最严重)

- Some banks came to view funding liquidity as essentially free, and funding liquidity risk as essentially zero.

2) Pooled "Average" Cost of Funds Approach 【一刀切】

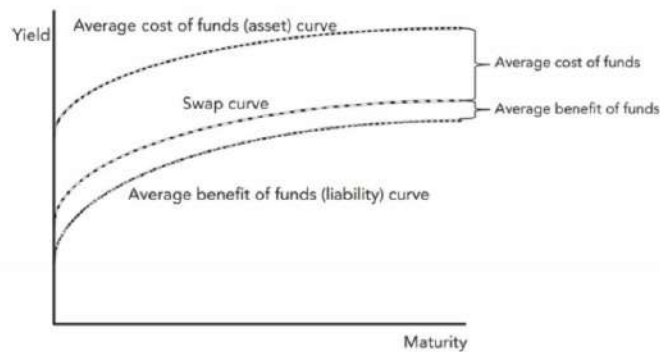
- An **average rate** was calculated based on the interest expense (cost of funds) across **all existing funding sources**.
- 根据所有现有资金来源的利息计算一个平均利率，所有资产无论其到期时间如何，收取相同的资金使用费。
- 例子：采用单一“平均”利率方法 Single average for the cost and benefit of funds
 - ✓ 假设所有资金来源的平均利率是 10BP。无论到期时间如何，1million 的本金，收取 1000 美元的费用。
 - ✓ 同样，无论到期时间如何，1million 的存款，获得 1000 美元的收益。

Term in years	1	2	3	4	5
Loan/deposit principal	\$1 million	\$1 million	\$1 million	\$1 million	\$1 million
Average cost of funds (bps)	10	10	10	10	10
Charge for use of funds	\$1,000	\$1,000	\$1,000	\$1,000	\$1,000
Credit for benefit of funds	\$1,000	\$1,000	\$1,000	\$1,000	\$1,000

● 缺陷:

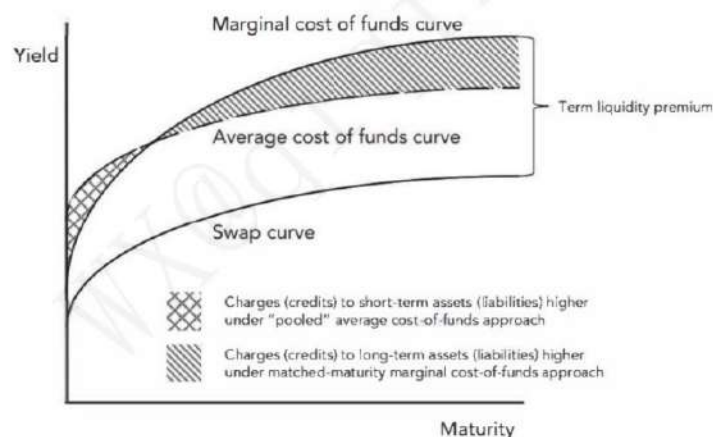
- ✓ First, it ignores the heightened 更高 **liquidity risk embedded in longer term assets**.
 - ✧ 采用单一“平均”利率法，忽略长期资产的流动性风险。假设所有资产无论其期限如何，都具有相同的流动性风险。
 - ✧ 改进方法：Separate Average Costs of Funds Approach
 - 例如下面这个例子：收取更高费用，支付更低的价格

Term in years	1	2	3	4	5
Loan/deposit principal	\$1 million	\$1 million	\$1 million	\$1 million	\$1 million
Average cost of funds (bps)	10	10	10	10	10
Average benefit of funds	4	4	4	4	4
Charge for use of funds	\$1,000	\$1,000	\$1,000	\$1,000	\$1,000
Credit for benefit of funds	\$400	\$400	\$400	\$400	\$400



- ✓ Second, if this "average" rate is also used to credit fund providers, then an incentive to write loans will be met with a direct disincentive to gather deposits.
 - ✧ Average Cost of Funds 反映的是历史价格，而不是当前的市场资金成本。当市场资金成本发生变化时，这种方法不能及时和准确地反映市场对新业务的风险感知。
- 影响：
 - ✓ Promotes Maturity Transformation.
 - ✧ Business units will be unduly encouraged to write long term assets because they do not receive higher charges for their use of funds over a longer period. 【因为没有额外的费用负担，因此资金使用部门可能更倾向于选择长期投资。】
 - ✧ Conversely, business units will be discouraged from raising long term liabilities because there is no premium credited to liabilities that provide funding for longer periods of time. 【因为没有获得额外的奖励，资金提供部门更不愿意筹集长期资金。】
 - ✧ 这种情况加剧了资产和负债期限的不匹配，增加了银行的流动性风险。
 - ✓ Distorts Profit Assessment 扭曲利润评估
 - ✧ The average cost of funds lags changes in banks' actual market cost of funds, especially in volatile markets.
 - ✧ 由于平均资金成本计算方法滞后于实际市场资金成本的变化。这种滞后效应会导致资产定价的不准确。
- 3) Matched-Maturity Marginal Cost of Funds Approach 匹配到期边际资金成本方法→目前最佳实践
 - A matched-maturity marginal cost of funds approach is the most accurate transfer pricing formula.

- 核心思想：From banks' actual market cost of funding, this approach calculates the **portion of the cost that is attributable to liquidity**.【基于实际市场资金成本，确定**流动性成本**】
- 找出流动性成本，就是找出 term liquidity premium(期限流动性溢价)
 - It seeks to achieve this by **converting fixed-rate borrowing costs to floating-rate borrowing costs** through an internal swap transaction and observing the spread over the reference rate, which is depicted from the swap curve. This spread is usually referred to as a **term liquidity premium**(**期限流动性溢价**) and is the rate that charges assets for the use of funds, and credits liabilities for the benefit of funds.
 - 做法：通过互换把 fixed-rate borrowing costs 转为 floating-rate borrowing costs。【固定利率借款成本包含多种因素，如信用风险、市场利率等，难以直接剥离出流动性成本。而转换为浮动利率后，与参考利率的利差能相对清晰地呈现出因流动性产生的成本或收益】
 - ❖ **term liquidity premium=floating-rate borrowing costs-reference rate(swap curve)**



- Under this approach, rates charged for the use of funds and, conversely, rates credited for the benefit of funds are based on the **term liquidity premiums** corresponding to the maturity of the transaction.
 - ✓ 对于长期资产，边际资金成本曲线(the marginal cost of funds curve)高于平均资金成本曲线(average cost of funds curve)。这反映了当前市场利率确定的 liquidity premium。
 - ✓ 对于短期资产，边际资金成本曲线低于平均资金成本曲线，反映了短期资产的较小期限流动性溢价。
 - ✓ 因此，LTP 的边际资金成本方法基于不同到期时间的市场利率。

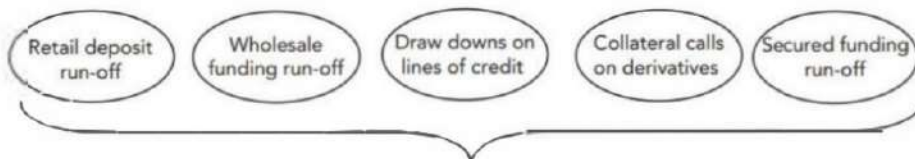
- Advantages:
 - ✓ First, it recognises that the costs and benefits of liquidity are related to the maturities of assets and liabilities, and therefore allows higher rates to be assigned to products that use or provide liquidity for longer periods of time. 流动性的成本和收益与资产和负债的到期时间相关。因此，对长期使用或提供流动性的产品分配更高的利率。
 - ✓ Second, it recognises the importance of having changes in market conditions incorporated quickly and efficiently into the rate used to charge and credit users and providers of funds, and therefore relies on the actual market cost of funds. 依赖实际市场资金成本，确保市场条件的变化能够快速且有效地反映在对应的利率中。
- 存款分类：存款分为“粘性 sticky”存款（如定期存款 Term Deposits）和“热/波动 Hot/volatile”存款（如活期存款、储蓄账户）。
 - ✓ 粘性存款不容易被提取，应该获得较大的激励 credit；
 - ✓ 热/波动存款更容易被提取，credit 应较低。

5. Contingent Liquidity Risk Pricing 或有流动性风险定价

- Calculating the charge for using, or the credit for providing, funding liquidity about **contingent commitments**. 针对潜在的流动性需求不确定性进行的成本定价。
- 例如，信用额度 lines of credit、衍生品的担保金和其他金融合同等在特定情况下要求银行提供额外的流动性
 - ✓ In these cases, the best approach is to impose a scenario model 银行通常采用情景模型预测各种可能的资金需求
 - ✓ Banks carry a **liquidity cushion**, or a "buffer" to help them survive periods of unexpected funding outflows.
- Process of Contingent liquidity risk pricing
 - ✓ Step 1: Identify contingent commitments that are likely to create unexpected funding demands. 识别那些可能会产生意外资金需求的或有承诺
 - ✓ Step 2: Perform stress tests under various scenarios to approximate the funding that might be required. 在不同情景下进行压力测试，估算可能需要的资金量
 - ✓ Step 3: Net approximations from above against inflows generated. 将估算的资金需求与生成的资金流入对比。

- ✓ Step 4: Calculate the cost of carry: the cost of funding liquid assets minus the return they generate. 计算持有成本
- ✓ Step 5: Recoup cost of carry: charging a liquidity premium, at the most granular level, to the business unit, product or transaction that creates the need for the bank to carry such liquid assets. 对业务部门、产品或交易收取流动性溢价，弥补持有流动资产的成本。

1. Identify contingent commitments that are likely to create unexpected funding demands.



2. Perform stress tests under various scenarios to approximate the funding that might be required.



3. Net approximations from above against inflows generated, for example, through the sale of marketable securities to derive the size of the liquidity cushion.

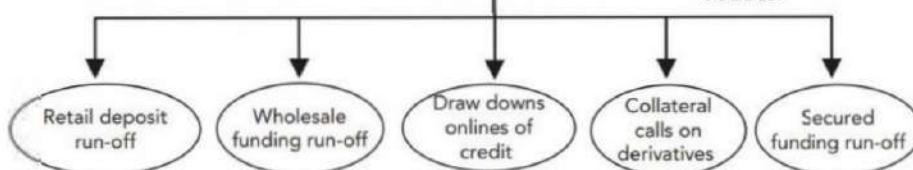
Liquidity cushion



Cost of carry

4. Calculate the cost of carry as the cost of funding liquid assets minus the return they generate. Ensure that appropriate haircuts and unsecured term funding charges have been applied to assets.

5. Recoup cost of carry by charging a liquidity premium, at the most granular level, to the business unit, product or transaction that creates the need for the bank to carry such liquid assets.



- Method for pricing contingent liquidity risk in credit lines
 - ✓ At the most basic level of what is considered to be better practice, all banks should be charging contingent commitments based on their likelihood of drawdown.
 - ✓ Rate charge for contingent liquidity risk = $(\text{limit} - \text{drawdown}) / \text{limit} \times \text{drawdown} \times \text{cost of liquidity cushion}$

例题：

A bank supplies a line of credit of \$10 million that currently has \$6 million already drawn. The bank determines that there is a 65% probability the customer will use the remaining line of credit. The bank's cost of funding for the liquidity cushion is 16 bps. If the bank charges contingent commitments based on the probability of a drawdown, what should the charge for liquidity be for this line of credit?

- A. \$1,664.
- B. \$2,496.
- C. \$4,160.
- D. \$5,850.

解答：C

Rate charge = $(10 - 6) / 10 \times 65\% \times 0.16\% = 0.0416\% = 4.16\text{BP}$

Multiplying this cost times the \$10 million credit line amount yields a dollar cost = $0.0416\% \times 10 \text{ million} = 4,160$.

Chapter 16 The US Dollar Shortage in Global Banking and the International Policy Response(☆)

美元短缺 (US Dollar Shortage) 指的是美元的需求超过了供应, 导致市场上美元变得稀缺。本节内容, 针对的是 2007-2009 年金融危机期间, 由于银行和金融机构对美元的需求激增, 而供应不足, 导致美元短缺的现象。

1. 全球金融危机期间美元短缺的原因

- 期限错配与融资缺口: The **maturity mismatch** between banks' U.S. dollar assets and liabilities and **rise in the funding gap** created short-term funding disruptions and made it difficult to obtain U.S. dollars through currency swaps. 银行的美元资产和负债之间存在期限错配, 加上融资缺口的增加, 银行难以获得足够的短期资金, 并且使用货币互换 currency swaps 获得美元变得更加困难和昂贵。
 - ✓ 期限错配 maturity mismatch: 存款到期时间短而贷款/投资到期时间长
 - ✓ funding gap 融资缺口: 短期资金需求与实际可获得的短期资金之间的差距
 - ✓ Currency Swaps: 金融危机期间, 由于市场动荡和对美元的需求增加, 使用货币互换 currency swaps 获得美元变得更加困难和昂贵
- 市场撤资:
 - ✓ withdrawal of money markets from bank-issued paper: 指的是投资者或机构不再购买或持有银行发行的短期金融工具
 - ✓ central banks withdrew a sizable portion of their U.S. dollar foreign exchange reserves: 中央银行减少了它们持有的美元外汇储备。撤回这些储备意味着市场上可用的美元供应减少了, 进一步加剧了美元短缺的问题。
- As U.S. dollar assets matured, banks needed to **roll over these assets**. Until mid-2008 banks accessed U.S. dollars directly in U.S. markets through their U.S. offices, including borrowing from the Fed. However, banks encountered difficulty in selling their less liquid U.S. dollar assets, including structured products, without incurring significant losses. **As a result, the holding period of assets lengthened while the funding maturity shortened, creating a U.S. dollar funding gap.** 银行在面临美元资产到期需要再融资时, 因流动性差的资产难以出售, 导致资产持有时间增加但融资到期时间缩短, 最终造成了美元融资缺口。

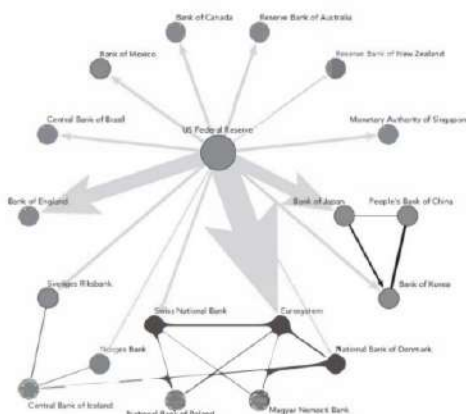
2. Maturity Mismatch&Currency Mismatch

- 期限错配 (Maturity Mismatch)：可能导致流动性风险。如果短期负债到期时，银行无法及时再融资，就可能面临资金链断裂的风险。这种情况在市场流动性紧张时尤为突出。
- 货币错配 (Currency Mismatch)：指金融机构的资产和负债以不同的货币计价。
 - ✓ 比如，银行持有大量以美元计价的资产，但其负债主要以其他货币（如欧元、日元等）计价，就会出现货币错配。
 - ✓ 如果汇率发生不利变化，银行可能会遭受汇兑损失。

3. The international policy response

【关于这一块知识，考纲要求是“Discuss how central bank swap agreements overcame challenges commonly associated with international lenders of last resort. 中央银行互换协议如何克服国际最后贷款人常见的挑战”】

- 小背景：
 - ✓ 中央银行互换协议 (Central Bank Swap Agreement) 指的是一个国家的中央银行与另一个国家的中央银行交换货币。
 - ✓ 通过中央银行互换协议，克服了传统国际最后贷款人 (international lenders of last resort)（如国际货币基金组织）在应对国际金融危机时常见的一些挑战(比如资金有限、响应速度慢、信息不对称等问题)。中央银行互换协议，使美联储和其他央行能够迅速、有效地提供所需的流动性，并且在此过程中避免了通常会出现的问题。
- 提供大量的美元流动性：由于各国央行缺乏足够的美元流动性来缓解其银行的资金短缺，通过与美联储的中央银行互换协议，这些央行能够获得所需的美元。
 - ✓ 最早的互换协议是美联储、欧洲央行和瑞士国家银行之间的协议，随后扩展到其他央行，形成了一个全球央行互换网络 (Central bank network of swap lines)。



- 随着 2008 年末危机的加剧，与几家主要欧洲央行的互换额度变为无限制，有效地将美联储变成了国际最后贷款人（Fed as an international lender of last resort）。无限制的互换额度确保了足够的流动性供应，减少了银行间利率的波动，并减轻了美元升值的压力。
- 通过互换网络 swap lines 进行贷款的央行不会面临信用风险（对手方违约），因为这些额度是有抵押的（外国货币）。这意味着美联储不直接承担最终借款人的信用风险，责任由各国央行分担，避免了信息不对称和道德风险的问题。
- swap lines 的使用也可以缓解其他货币的压力。

Chapter 17 Covered Interest Parity Lost 利率平价理论的失衡(☆)

本节探讨了一个问题：自 2007-2009 年金融危机以来，为什么抛补利率平价 (covered interest parity, CIP) 的持续存在偏离 (即 CIP 不成立)？因为作为经典的金融利率，若 CIP 不成立，套利者可以利用套利机会，使得 CIP 始终成立。

1. 基本概念：Covered Interest Parity(CIP)抛补利率平价【一级知识】

- ✓ no-arbitrage condition according to which interest rates on two otherwise identical assets in two different currencies should be equal once the foreign currency risk is hedged:

$$F = S \times \frac{(1 + R_{YYY})^T}{(1 + R_{XXX})^T}$$

$$\frac{F}{S} = \frac{1 + r}{1 + r^*}$$

- ✧ S is the **spot exchange rate**(现货汇率/即期汇率) in units of US dollar per foreign currency
- ✧ F is the corresponding **forward exchange rate**(远期汇率) 【注意汇率的表达方式】
- ✧ r is the US dollar interest rate
- ✧ r* is the foreign currency interest rate.
- ✧ T 指的是时间，这里约定 T=1year
- ✓ 如果实际市场中 F 和 S 不满足这个关系，就意味着 CIP 不成立，这就是所谓的“CIP Lost”。

2. FX Swaps and Cross-currency Swaps

● FX Swaps:

- ✓ In an **FX swap**, one party borrows one currency from, and simultaneously lends another currency to, a second party. 一方从另一方借入一种货币，同时借出另一种货币
 - The borrowed amounts are exchanged at the spot rate and then repaid at the **pre-agreed forward rate** at maturity. 借入的金额按现货汇率(S)交换，然后在到期时按预先约定的远期汇率(F)偿还。
 - 期限短；风险小
- ✓ 如果 CIP 成立，那么通过 FX swap 借入某种货币的成本应该与直接在现金市场借入该货币的成本一致。
 - 但是，实际中，尤其是在全球金融危机中，这种平价关系被打破，因为市场对美元的需求较高，通常需要支付额外的费用或利率差。

$$F - S > S \times \left(\frac{1+r}{1+r^*} - 1 \right) \rightarrow \text{CIP 不成立}$$

forward points : (F - S)

● **Cross-currency swap** 【交叉货币互换，就是一级学的货币互换】

- ✓ 在合同开始时按现货汇率(S)交换货币，在合同期限内定期交换利息支付，并在到期时按**初始现货汇率(S)**交换货币。【不同的货币互换对于最后本金互换时的汇率选择不一样，有的是远期汇率，注意我们这一节，用的是现货汇率】

- 期限长，通常大于 1 年

- ✓ CIP 的失效会导致借入某种货币(美元)的成本上升。借入美元的一方可能需要支付高于现金市场利率的基差(cross-currency swap basis/basis,b)

$$F - S = S \times \left(\frac{1 + r + b}{1 + r^*} - 1 \right)$$

- 当 b 不为零时，表明抛补利率平价 (CIP) 被违反。

- 实际上，自 2007 年以来，以美元借贷与大多数货币（如欧元和日元）的基差为负，意味着通过外汇互换市场借入美元比在美元现金市场直接融资更昂贵；而对于一些货币（如澳元），基差为正。

- ✓ 影响 **cross-currency swap basis** 的要素

- ◇ decline in market liquidity

- Market liquidity in the underlying instruments may evaporate, so that the difference between bid and ask prices for forward and spot transactions is non-trivial.

- ◇ Risk increases of the underlying instruments

- For example, a rise in counterparty credit risks in the interbank markets

3. **Causes of CIP Violations** 【全球金融危机后，CIP 仍然不成立，也就是 cross-currency swap basis 持续存在的原因】

- ✓ Arbitrage become **both costly and risky after crisis**. 套利成本高且风险大【核心：套利者需承担资本占用、市场风险、信用风险、交易对手风险和流动性风险等成本】

- Emerging market with capital restriction

- As a result of tighter management of risks and related balance sheet constraints, arbitrage now incurs a cost per unit of balance sheet. 更严格的风险管理和相关的资产负债表约束

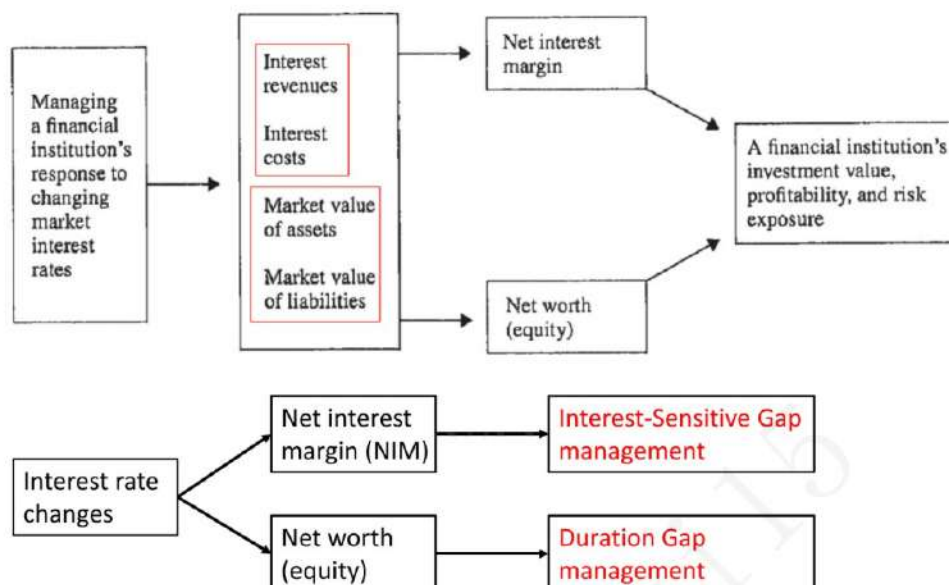
- ◆ Balance sheet deleverage

- This cost is passed on to the pricing of FX swaps, introducing a premium.
这种成本被转嫁到 FX swaps 的定价中，引入了溢价
- ✓ **Changes in regulation** have reinforced market pressures for a tighter management of balance sheet risks. 【监管变化(如 Basel III 资本要求)限制了套利活动】

Chapter 18 Risk Management for Changing Interest Rates(☆☆☆)

这一节的内容，在有的教材中，也放在市场风险中。主要涉及利率变化对于银行 profit 的影响。

1. 基本概念



2. Interest-sensitive Gap Management

- 有的教材也叫 Repricing model 或者 funding gap model
- 管理的目标：利率波动不影响银行的 **profits**
 - ✓ 即 net interest margin (NIM 净利差)固定 → NII 不波动 → $\Delta NII=0$
- 银行资产/负债分类：
 - 1) 可调整利率的（利率敏感的）资产/Repriceable (Interest-Sensitive) Assets
 - **Interest-Sensitive Assets(ISA)**利率敏感性资产
 - ✓ 利率发生变化会影响其收益。例如：
 - ✧ Short-term securities issued by governments and private borrowers (about to mature 即将到期)
 - ✧ Short-term loans made to borrowing customers (about to mature 即将到期)
 - ✧ Variable-rate loans and securities
 - 2) 可调整利率的（利率敏感的）负债 Repriceable (Interest-Sensitive) Liabilities
 - **Interest-Sensitive Liabilities(ISL)**利率敏感性负债
 - ✓ 利率发生变化会影响其支出。例如：
 - ✧ Borrowings from the money market (such as federal funds or RP borrowings)
 - ✧ Short-term savings accounts

- ✧ Money-market deposits (whose interest rates are adjustable frequently)
- ✓ 【关于 ISA/ISL, 有的教材也写作 RSA/RAL(R:rate)】
- 3) Nonrepriceable Assets 不可重定价资产
 - ✓ 利率变化对其价值没有影响。例如:
 - ✧ Cash in the vault and deposits at the Central Bank (legal reserves)
 - ✧ Long-term loans made at a fixed interest rate
 - ✧ Long-term securities carrying fixed rates
 - ✧ Buildings and equipment
- 4) Nonrepriceable Liabilities 不可重定价负债
 - ✓ 利率变化对其价值没有影响。例如:
 - ✧ Demand deposits (which pay no interest rate or a fixed interest rate)
 - ✧ Long-term savings and retirement accounts
 - ✧ Equity capital provided by the financial institution's owners
- 重要公式:

★ $NIM = NII / \text{earning assets}$
★ $NII = \text{interest income} - \text{interest expense}$
★ 简化公式 $\rightarrow NII = \text{interest income} - \text{interest expense} = ISA \cdot i_{ISA} - ISL \cdot i_{ISL}$
★ $\Delta NII = ISA \cdot \Delta i_{ISA} - ISL \cdot \Delta i_{ISL} = (ISA - ISL) \cdot \Delta i$ 【假设 $\Delta i_{ISA} = \Delta i_{ISL} = \Delta i$, i : 利率】
★ $\text{Interest-sensitive gap (IS gap)} = ISA - ISL$
★ $\text{cumulative gap} = \text{summing the IS gaps}$
★ $\text{Relative IS gap} = \text{IS gap} / \text{Size of financial institution}$
★ $\text{Interest Sensitivity Ratio (ISR)} = ISA / ISL$

- IS gap
 - ✓ $IS\ gap = 0$: 利率变化不影响 NII 和 NIM.
 - ✓ Positive IS GAP $\rightarrow IS\ gap > 0$: 利率上升(下降) $\rightarrow$ NII 和 NIM 上升(下降);
 - ✓ Negative IS GAP $\rightarrow IS\ gap < 0$: 利率上升(下降) $\rightarrow$ NII 和 NIM 下降(上升).
- 估算 IS gap 的步骤:
 - 1) Management must **choose the time period** during which the net interest margin (NIM) is to be managed (e.g., six months or one year) to achieve some desired value and the length of subperiods ("**maturity buckets**") into which the planning period is to be divided.

- ✓ maturity buckets: 特定的时间区间, 用于分类和计算资产和负债在这些时间段内的重新定价情况。例如:
 - ✧ 0 to < 1 month
 - ✧ more than 1 month, up to 3 months
 - ✧ more than 3 months, up to 6 months
 - ✧ more than 6 months, up to 12 months
 - ✧ more than 1 year < 2 years
- 2) Management must choose a **target level for the net interest margin**- that is, whether to freeze the margin roughly where it is or perhaps increase the NIM. 【确定 NIM 目标, 让 NIM 上升还是不变】
- 3) If management wishes to increase the NIM, it must either develop a **correct interest rate forecast** or find ways to reallocate earning assets and liabilities to increase the spread between interest revenues and interest expenses. 【正确的预测利率】
- 4) Management must **determine the volume of interest sensitive assets and interest-sensitive liabilities** it wants the financial firm to hold. 【确定 ISA 和 ISL 的量】
- **IS gap 管理方法**
 - ✓ ① Defensive gap management 防御缺口管理
 - ✧ The goal of defensive gap management is to **set interest-sensitive GAP as close to zero** as possible to reduce the expected volatility of net interest income.
 - ✧ 令 IS **GAP=0**, 减少净利息收入的预期波动
 - ✓ ② Aggressive gap management 激进缺口管理:

预测利率变化		积极管理最可能的行动
Rising market interest rates 市场利率上升	Positive IS GAP	Increase interest-sensitive assets 增加利率敏感性资产
		Decrease interest-sensitive liabilities 减少利率敏感性负债
Falling market interest rates 市场利率下降	Negative IS GAP	Decrease interest-sensitive assets 减少利率敏感性资产
		Increase interest-sensitive liabilities 增加利率敏感性负债
- **IS GAP Management 问题**
 - ✓ Changes inconsistent 利率变动不一致: Interest rates paid on liabilities tend to move faster than interest rates earned on assets

- ✓ Basis risk 基差风险: The interest rates attached to assets of various kinds often change by different amounts and at different speeds than many of the interest rates attached to liabilities.

- ✓ Net worth is not considered.

3. Duration Gap Management

- 对于大多数公司来说, 净值(资产和负债本身价值)比净利息收益(利息收益和利息费用)更重要, 而 IS GAP 管理未考虑净值。
- The net worth (NW) of any business or household is equal to the value of its assets less the value of its liabilities:

$$NW = \text{assets} - \text{liabilities} = A - L$$

$$\Delta NW = \Delta A - \Delta L = \left(-D_A \times \frac{\Delta i}{1+i} \times A \right) - \left(-D_L \times \frac{\Delta i}{1+i} \times L \right)$$

$$= - \left(D_A - D_L \times \frac{L}{A} \right) \times \frac{\Delta i}{1+i} \times A$$

✧ D_A : 资产的麦考林久期; D_L : 负债的麦考林久期

✧ $D_A - D_L$: Duration gap

✧ $D_A - D_L \times \frac{L}{A}$: Leverage-adjusted duration gap

✧ $\frac{L}{A}$: leverage adjusted indicator

一级知识回顾: 麦考林久期(D)和修正久期(MD)

- Macaulay Duration 麦考林久期(D)
- ✓ The weighted average time to receipt of the bond's promised payments.
- ✓ 未来现金流的平均回收期限
- ✓ 麦考林久期本质上是时间

$$D_{\text{Macaulay}} = \sum_{t=1}^T \left[\frac{PV(C_t)}{P} \times t \right] = \sum_{t=1}^T (\omega_t \times t)$$

- 债券价格变化(ΔP) $= -MD \times \Delta i \times P = -\frac{D}{1+i} \times \Delta i \times P$

● 分析

Leverage-adjusted duration gap	interest	Net Worth	Value change of Asset/Liability
Positive	Rise	Decrease	Assets value ↓ ↓ Liabilities value ↓
	Fall	Increase	Assets value ↑ ↑ Liabilities value ↑
Negative	Rise	Increase	Assets value ↓ Liabilities value ↓ ↓
	Fall	Decrease	Assets value ↑ Liabilities value ↑ ↑
Zero	Rise	No change	Assets value ↓ Liabilities value ↓
	Fall	No change	Assets value ↑ Liabilities value ↑

● Duration gap management

① Defensive gap management

Portfolio immunization strategy: $D_A - D_L \times \frac{L}{A} = 0$

② Aggressive gap management

Expected Change in Interest Rates	Aggressive Management's Most Likely Action	Best Interest-Sensitive GAP Position to Be in
Rates $\uparrow$	$D_A - D_L \times \frac{L}{A} < 0$: Reduce DA and increase DL	Net worth increase
Rates $\downarrow$	$D_A - D_L \times \frac{L}{A} > 0$: Increase DA and reduce DL	Net worth increase

● Limitations

- ✓ Finding assets and liabilities of the same duration that fit into a financial-service institution's portfolio is often a frustrating task. 【找到具有相同久期的资产和负债非常困难】
- ✓ Duration gap analysis tends to be reasonably effective at handling interest rate risk problems if the yield curve (i.e., the maturity structure of interest rates) changes by relatively small amounts and moves in parallel steps with short term and long-term interest rates changing by about the same proportion over time. 【久期缺口分析在利率小幅度、平行变动时相对有效】
- ✓ Add convexity concept into calculation to get more precise result for large interest rate movements. 【引入凸度更好，但是这里没有考虑凸度问题】
- ✓ Duration itself can shift as market interest rates move. 【久期本身会随着市场利率的变化而变化】

Chapter 19 Illiquid Assets 非流动性资产(☆☆)

非流动性资产 (Illiquid Assets) 是指在短时间内难以迅速变现的资产，流动性风险相对较大。注意，在一定市场情况下，流动性好的资产也会变得缺乏流动性。反之亦然。

1. Illiquid Asset Markets

● Characteristics

- ✓ Most Asset Classes Are Illiquid 大多数资产类别都是非流动性的
- ✓ Illiquid Asset Markets Are Large
- ✓ Investors Hold Lots of Illiquid Assets
- ✓ Liquidity Dries Up 流动性枯竭：流动性危机经常发生，因为在市场严重压力期间，流动性往往会枯竭

2. The Source of Illiquidity Market Imperfections

● Clientele Effects and Participation Costs 客户效应和参与成本：

- ✓ 投资者需要花金钱、时间或精力来学习和掌握必要的技能。

● Transaction Costs 交易成本：

- ✓ 包括佣金、税费和某些流动性差的资产的尽职调查、产权转让等成本，以及常规交易成本。

● Search Frictions 搜索摩擦：

- ✓ 许多资产需要寻找合适的买家或卖家，可能需要很长时间才能完成交易。

● Asymmetric Information 信息不对称

● Price Impact 价格影响：大额交易会影响市场价格。

● Funding Constraints 资金限制：如果信贷市场紧缩，投资者很难在流动性差的市场交易，因为这些投资通常高度依赖杠杆。

3. Biases on Returns for Illiquid Assets

● 由于下面 3 个偏差的出现，使得非流动资产的收益被高估，风险被低估。

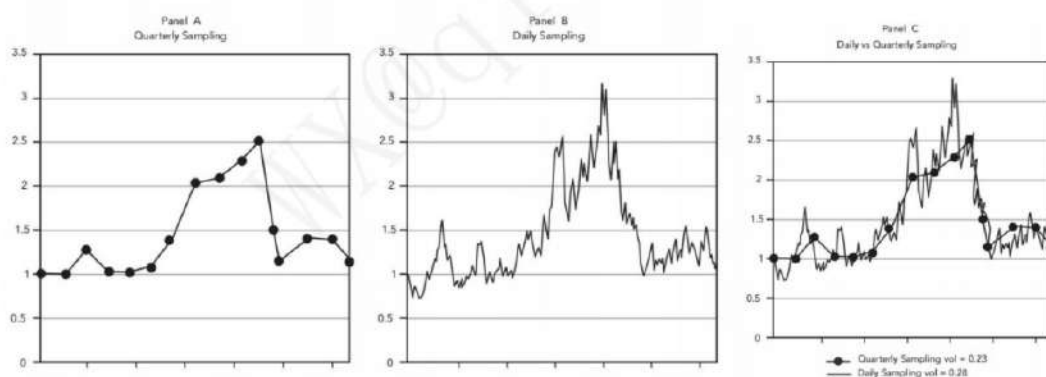
● 前两个偏差可以结合《投资》科目的“Chapter 9 Hedge Fund 对冲基金”一起理解

1) Survivorship bias 生存偏差 【高估收益】

- 由于仅观察和分析那些在某段时间内存活下来的样本，而忽略了那些已经退出或失败的样本，从而导致结果偏向于存活者的现象表现差的资产收益，没有被统计。
- Survivorship bias results from the tendency of poorly performing funds to stop reporting. Many of these funds ultimately fail—but we only rarely count

their failures. This makes true illiquid asset returns worse than the reported data.

- 除了生存偏差外，还存在其他数据偏差-Reporting bias 报告偏差
 - Reporting bias 是指由于报告收益是自愿行为，会导致数据不准确。一种情况是基金业绩不佳，一开始就不报告收益；另一种是基金原本在数据库中，但预期收益低就停止报告。
- 2) **Selection bias** 选择偏差【高估收益，低估风险】
 - 选择偏差是由于数据样本的选择方式存在偏差，使得样本不能代表总体，从而导致分析结果不准确的现象。
 - Sample selection bias results from the tendency of returns only to be observed when underlying asset values are high, and result in underestimated risk as measured by beta and volatility
 - 生存偏差是由于仅分析存活者数据而忽略了失败者数据导致偏差，选择偏差是由于数据选择方式存在系统性偏差导致的结果偏差。
- 3) **Infrequent Trading** 不频繁交易 【低估风险】
 - With infrequent trading, estimates of risk—volatilities, correlations, and betas—are too low when computed using reported returns. Thus risk estimates are biased downward by infrequent sampling



- **Unsmoothing** 去平滑
 - ✓ 为了弥补不频繁交易的偏差，需要进行“去平滑 Unsmoothing”处理，即从季度数据中提取出更真实、更波动的日常数据。【由简至繁】
 - ✓ Unsmoothing is a filter problem 滤波问题。
 - ✧ 简单来说，就是通过人为地增加波动性，使数据更接近实际情况。
 - ✓ Unsmoothing adds noise back to reported returns to uncover the true, noisier returns.

- ✓ This process affects risk and return estimates and could have a dramatic effect on returns.
- ✓ **Properties**
 - ✧ Unsmoothing affects **only risk estimates and not expected returns**. 去平滑只影响风险估计，不影响预期回报
 - ✧ Unsmoothing has no effect if the observed returns are uncorrelated.
 - ✧ Unsmoothing is an art.
- 4. 获得流动性风险溢价的方法
 - 1) **Holding passive allocations** 被动配置 **to illiquid asset classes**
 - 2) **Holding less liquid securities within asset classes**
 - 3) Market making at the individual security level
 - 4) **Dynamic rebalancing** at the aggregate level: **the simplest** to implement and has the **greatest impact** on portfolio returns.
- Across asset classes 和 Within asset classes
 - ✓ Across asset classes 【即跨资产类别的流动性风险溢价并不明显】
 - ✧ 尽管传统观点认为投资者在承受流动性不足风险时可以获得流动性风险溢价，但这些观点并不完全可靠，原因如下：
 - Illiquidity biases. (**overestimated asset returns**)
 - Much **ignored risk**. (Illiquid asset classes contain far more than just illiquidity risk)
 - There is no “market index” for illiquid asset classes. 投资者无法获得非流动性资产类别的市场指数回报
 - You cannot separate factor risk from manager skill. (investing in illiquid markets is always a bet on management talent.)
 - ✓ Within asset classes 【即具体资产类别内的流动性溢价较大】
 - ✧ Within all the major asset classes, these illiquidity premiums can be accessed by dynamic factor strategies which take long positions in illiquid assets and short positions in liquid ones.
 - U.S. Treasury bond: on-the-run/off-the-run
 - Corporate bond: one that trades less frequently or has larger bid-ask spreads has higher returns
 - Equity: bid-ask spreads, volume, turnover

- Illiquidity asset: Hedge funds that place more restrictions on the withdrawal of capital
- ✓ Large illiquidity risk premiums exist **within asset classes** but not across asset classes 【在具体资产类别内更容易获取流动性风险溢价】

5. Portfolio Choice Decisions

- 包含非流动性资产的投资组合:
 - ✓ 应仅持有适量的非流动性资产;
 - ✓ 并要求较高的非流动性风险溢价;
 - ✓ 投资者需要考虑个人情况、交易成本、流动性事件频率以及资产配置对消费和再平衡策略的影响来做出决策。